nature synthesis

Article

https://doi.org/10.1038/s44160-024-00662-x

Carbon- and energy-efficient ethanol electrosynthesis via interfacial cation enrichment

Received: 30 August 2023

Accepted: 6 September 2024

Published online: 4 October 2024

Check for updates

Ali Shayesteh Zeraati[1,5], Feng Li[1,5], Tartela Alkayyali[1,5], Roham Dorakhan[2,5], Erfan Shirzadi[2], Fatemeh Arabyarmohammadi[1], Colin P. O'Brien[1], Christine M. Gabardo[1], Jonathan Kong[3], Adnan Ozden[1,4], Mohammad Zargartalebi[1], Yong Zhao[1], Lizhou Fan[2], Panagiotis Papangelakis[1], Dongha Kim[2], Sungjin Park[2], Rui Kai Miao[1], Jonathan P. Edwards[1], Daniel Young[1], Alexander H. Ip[2], Edward H. Sargent[2] & David Sinton[1]

The use of acidic electrolytes in CO_2 reduction avoids costly carbonate loss. However, the energy efficiency of acid-fed electrolysers has been limited by high hydrogen production and operating potentials. We find that these stem from the lack of alkali cations at the catalyst surface, limiting CO_2 and CO adsorption. In acid-fed membrane electrode assembly systems, the incorporation of these cations is challenging as there is no flowing catholyte. Here an interfacial cation matrix (ICM)–catalyst heterojunction is designed that directly attaches to the catalyst layer. The negatively charged nature of the ICM enriches the alkali cation concentration near the cathode surface, trapping generated hydroxide ions. This increases the local electric field and pH, increasing multi-carbon production. Integrating the ICM strategy with a tailored copper–silver catalyst enables selective ethanol production through a proton-spillover mechanism. We report a 45% CO_2-to-ethanol Faradaic efficiency at 200 mA cm^{-2}, carbon efficiency of 63%, full-cell ethanol energy efficiency of 15% (3-fold improvement over the best previous acidic CO_2 reduction value) and energy cost of 260 GJ per tonne ethanol, the lowest among reported ethanol-producing CO_2 electrolysers.

Electrochemical CO_2 reduction (CO_2R) can utilize CO_2 emissions and produce fuels via renewable electricity. Recent research has resulted in many advancements to catalyst design, product selectivity and electrolyser configuration[1–4]. Ethanol is of particular interest among reported CO_2R products owing to its high energy density and use as a fuel. As ethanol is a platform chemical with a global market size exceeding \$100 billion yr^{-1}, the decarbonization of ethanol production is a priority[5]. Biomass-based ethanol production has solidified it as a platform chemical and fuel, but the associated carbon intensity remains high at >1.7 kg CO_2 equivalents per kilogram of ethanol (Fig. 1a). The electroreduction of CO_2 offers a promising pathway for reducing the carbon intensity associated with ethanol production, with profitability being reliant on high-performance metrics. Most advancements in ethanol electrosynthesis using CO_2R have been achieved in alkaline and neutral electrolytes[6–12]. These electrolytes facilitate an alkaline environment in the catalyst layer[13], which provides favourable conditions for

[1]Department of Mechanical and Industrial Engineering, University of Toronto, Toronto, Ontario, Canada. [2]Department of Electrical and Computer Engineering, University of Toronto, Toronto, Ontario, Canada. [3]Department of Materials Science and Engineering, University of Toronto, Toronto, Ontario, Canada. [4]Department of Mechanical and Nuclear Engineering, Khalifa University, Abu Dhabi, United Arab Emirates. [5]These authors contributed equally: Ali Shayesteh Zeraati, Feng Li, Tartela Alkayyali, Roham Dorakhan. e-mail: sinton@mie.utoronto.ca

https://doi.org/10.1038/s44160-024-00662-x

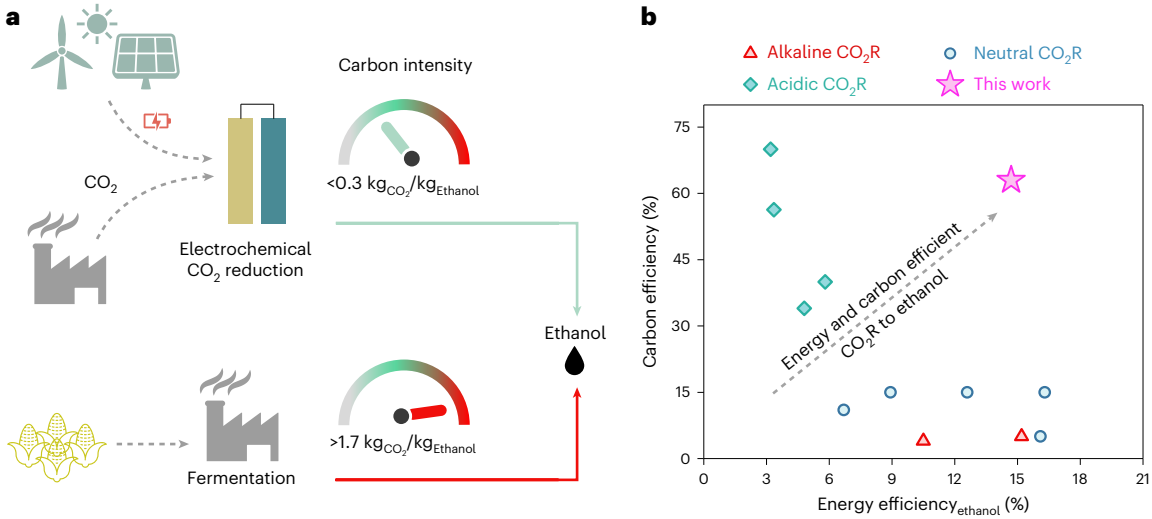


Fig. 1 | Carbon intensity and performance evaluation of CO_2R in MEA electrolysers. a, The carbon intensity for ethanol production from CO_2R and biomass[56]. The carbon intensity for the CO_2R route was calculated on the basis of the carbon intensity of renewable electricity used by the electrolyser. **b**, A comparison of carbon and energy efficiency for ethanol electrosynthesis in the benchmark alkaline, neutral and acidic CO_2 MEA electrolysers. The results are based on systems operating at current densities ≥100 mA cm^{-2}. Alkaline MEA electrolysers are reported in refs. 9,10, neutral MEA electrolysers are reported in refs. 6–8,11,12 and acidic MEA electrolysers are reported in refs. 16–18,40.

CO_2R to multi-carbon (C_{2+}) products with ethanol energy efficiencies ($EE_{ethanol}$) approaching 10% (Fig. 1b). However, the CO_2R evolves hydroxide ions (OH^-) that then combine with unreacted, dissolved CO_2 to form negatively charged (bi)carbonate ions (Supplementary Note 1 and Supplementary Fig. 1). This (bi)carbonate ion formation limits the carbon efficiency (defined as the percentage of CO_2 converted per total CO_2 input; equivalent to single-pass carbon efficiency (SPCE)) to typically below 20% towards multi-carbon products, resulting in a severe CO_2 recovery cost penalty[14,15]. Operation of CO_2R in acidic conditions can alleviate this (bi)carbonate formation issue. In acidic conditions, (bi)carbonate ion formation is mitigated by the high proton concentration in the electrolyte surrounding the cathode, enabling higher carbon efficiency (>30%)[16–18]. However, the unshielded cathode surface substantially favours the hydrogen evolution reaction (HER) over CO_2R owing to the high availability of protons and the fast kinetics of the HER, prohibiting high $EE_{ethanol}$ (Fig. 1b)[16,18]. This compromise between energy efficiency and carbon efficiency is fundamental to current alkaline, neutral and acidic approaches and motivates material and system design principles that would enable ethanol electrosynthesis with practical carbon efficiency and $EE_{ethanol}$.

To achieve a high $EE_{ethanol}$, we initially sought to increase the CO_2R Faradaic efficiency (FE) towards ethanol at a high current density and a low full-cell voltage in a carbon-efficient electrolyser. Membrane electrode assembly (MEA) electrolysers demonstrate lower full-cell voltages than their flow cell counterpart and offer a path towards industrial scalability. Operating CO_2R in an MEA with an acidic electrolyte and a cation-exchange membrane (CEM; Supplementary Fig. 1b) leads to ~100% HER and zero CO_2 crossover[16,17]. We recently demonstrated porous proton-blocking ad-layer strategies, enabling acidic CO_2R in flow cell electrolysers[19,20]. However, that strategy failed to translate to the MEA system, with the HER dominating all other reactions (Supplementary Fig. 2). The very different reaction environment present in the MEA demands a new approach[16,17,19,20].

In this study, we aimed to produce ethanol via CO_2 electrocatalysis with high carbon efficiency and $EE_{ethanol}$. In a forward-bias bipolar membrane (f-BPM) MEA operating with an acidic anolyte, we found that the lack of alkali cations (such as K^+) at the cathode surface limits CO_2 or CO adsorption on Cu, which leads to high HER and low $EE_{ethanol}$. An additional cause of reduced performance was hydroxide ion migration from the cathode surface to the anion-exchange membrane (AEM)–CEM interface of the BPM, reducing alkalinity on the cathode and, therefore, diminishing C–C coupling (a key step along the pathway to C_{2+} products). To overcome these challenges, we developed an interfacial cation matrix (ICM)–catalyst heterojunction to enrich alkali cations at the catalyst surface and enable trapping of locally generated OH^-, resulting in selectivity for CO_2R products at an FE_{CO_2R} of ≥90% and HER FE of ≤10% at 200 mA cm^{-2}. We combined the ICM strategy with a Cu–Ag catalyst designed for ethanol electroproduction, resulting in simultaneous achievement of 63% carbon efficiency, 15% full-cell $EE_{ethanol}$ and a total energy cost of 260 GJ per tonne ethanol.

Results and discussion

Local CO_2R environment under acidic conditions

To assess the catalyst surface conditions when CO_2R is operated in an acid-fed MEA, we employed finite element method (FEM) simulation (Supplementary Fig. 3). If no AEM was used, the local catalyst environment would be expected to be highly acidic due to proton accumulation at the outer Helmholtz plane (OHP), resulting in 100% HER (Supplementary Fig. 1d). Depositing an anion-exchange layer (AEL) on the catalyst surface would increase the local pH, with thicker layers favouring CO_2R (Supplementary Fig. 3). The use of an AEL maintains a local alkaline environment in acidic CO_2R, but it fails to completely shield the cathode from proton electromigration, leading to a high H_2 FE particularly at high-carbon-efficiency regimes[16,17,19]. We also found that the proton accumulation at the OHP results in a low electric field intensity (Fig. 2a), which could likewise favour HER. Therefore, we studied the effect of adding a salt to the bulk electrolyte. Alkali cations from the salt would electromigrate to the cathode and accumulate at the OHP (Fig. 2b), increasing the local electric field intensity compared with that produced when only protons are present (Fig. 2a)[21]. Previous research also showed better CO_2R selectivity as the local cathodic electric field intensity increased[22,23]. Therefore, using alkali cations in f-BPM CO_2 electrolysers could suppress the competing HER by increasing the FE towards multi-carbon products. However, high concentrations of cations need to be added to the anolyte to improve CO_2R FE, which may lead to salt precipitation and stability issues[9,24–26]. The FEM simulation demonstrated that even a low concentration of cations at the cathode surface could result in a sufficient electrical field for CO_2R

https://doi.org/10.1038/s44160-024-00662-x

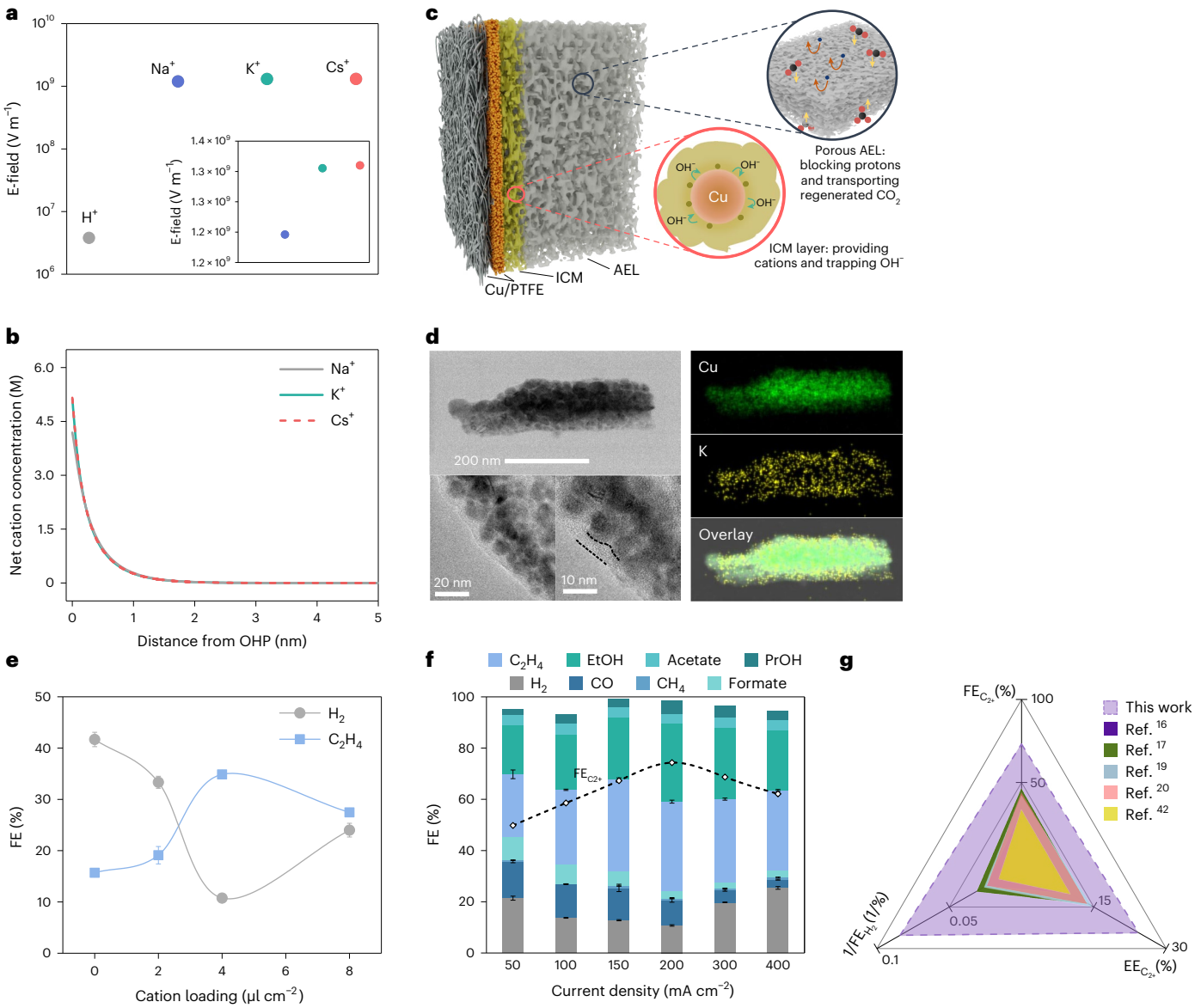


Fig. 2 | Characterization of the ICM with a Cu catalyst. a,**b**, Modelled electric field (E-field) intensity (**a**) and net cation concentration (**b**) of different cations under an applied current density of 100 mA cm^{-2}. **c**, GDE equipped with an ICM and a porous AEL. (i) The ICM provides cations close to the cathode surface and traps the generated OH$^-$; (ii) the AEL blocks the proton flux from the anode and transports the generated CO_2 to the cathode surface for further reduction. **d**, STEM with the corresponding elemental EDS mapping of K and Cu and TEM images of ICM–Cu on a PTFE nanofibre. These show a conformal coating of ICM and the presence of potassium on the Cu surface. **e**, Hydrogen and ethylene FE at different concentrations of potassium in the ICM at 200 mA cm^{-2}. The case where there is zero cation loading indicates the absence of ICM. **f**, The FE of CO_2R products and H_2 at different current densities with the corresponding C_{2+} FE (composed of ethylene, ethanol (EtOH), *n*-propanol (PrOH) and acetate) in a Cu-based MEA equipped with an ICM and 0.01 M H_2SO_4 anolyte. **g**, A comparison of FE$_{C_{2+}}$, C_{2+} energy efficiency (EE$_{C_{2+}}$) and H_2 FE of an ICM-equipped Cu GDE with the performance metrics of previously reported CO_2R electrolysers. Comparisons are constrained to reports with carbon efficiency of ≥60% (refs. 16,17,19,20,42). For **e** and **f**, the plotted values are means of three replicates, while error bars indicate the standard deviation.

(Supplementary Fig. 4). To investigate the effect of a low concentration of alkali cations on CO_2R activity, we performed surface-enhanced Raman spectroscopy measurements on Cu electrodes in contact with 0.01 M H_2SO_4 and 0.01 M H_2SO_4 + 5 mM of K_2SO_4 electrolytes in a flow cell (Supplementary Note 2 and Supplementary Figs 5 and 6). The results show that the addition of a low-concentration cation source facilitates CO_2R. By considering this observation, we sought a way to have cations present locally but not in the bulk electrolyte in an acid-fed MEA.

One approach would be to enrich the catalyst surface with cations with a multifunctional heterojunction. Perfluorinated sulfonic acid (PFSA) ionomers, such as Nafion, consist of a hydrophobic polytetrafluoroethylene (PTFE, –CF_2) backbone and a hydrophilic perfluorinated vinyl ether side-chain terminated with a sulfonate group (–SO_3^-)[27,28]. The sulfonate groups are neutralized by another cation, often a proton, leading to the formation of sulfonic acid groups (–SO_3H)[28–30]. By spray coating a PFSA ionomer solution on the cathode (sputtered Cu on a porous PTFE gas diffusion electrode (GDE)), the –SO_3H group is expected to be preferentially exposed to the hydrophilic catalyst surface. We reasoned that exchanging the protons on the PFSA ionomers with alkali cations (thereby creating the ICM) would tailor the local microenvironment and provide a sufficiently alkali cation-rich environment at the OHP (Fig. 2c). ICM alters local pH by excluding anions and accumulating CO_2R-generated OH$^-$ at the catalyst surface. The negatively charged sulfonate groups in ICM inhibit OH$^-$ transport

https://doi.org/10.1038/s44160-024-00662-x

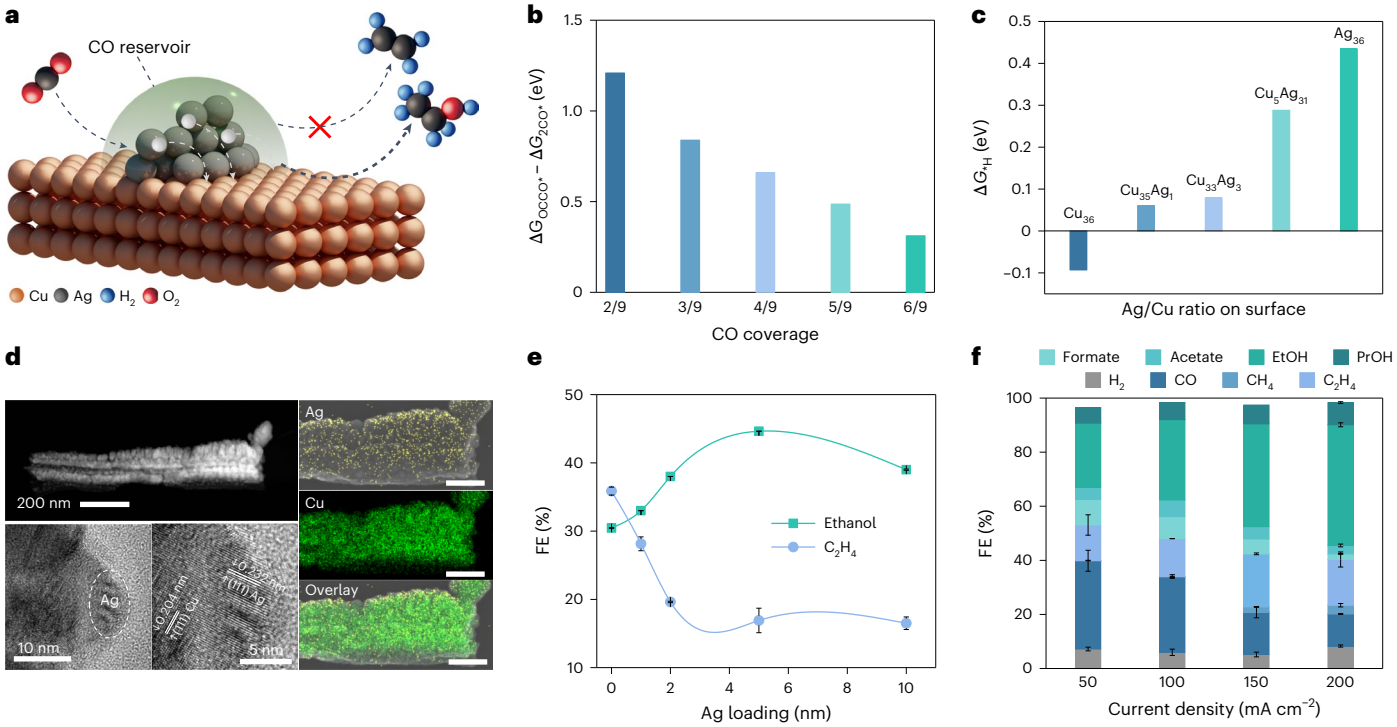


Fig. 3 | Design of a Cu–Ag catalyst for enhanced ethanol production. **a**, A schematic of a Cu–Ag catalyst where Ag clusters (grey) on the Cu surface (orange) can aid in increasing the local concentration of CO, while weakening H* binding (white) and lowering HER activity. **b**, The C–C coupling free energy of Cu under different CO* coverages. **c**, The hydrogen binding energy on Cu, Ag and CuAg alloys at different Ag contents. **d**, STEM with the corresponding EDS elemental mapping of Ag and Cu as well as a HRTEM image of Cu–Ag on a PTFE nanofibre. The loading of Ag is 5 nm. The scale bar in EDS images represents 100 nm. **e**, Ethanol and ethylene FE at different loadings (measured through sputter deposition thickness) of Ag on top of a Cu GDE. The data were obtained at a current density of 200 mA cm^{-2} and an anolyte composition of 0.01 M H_2SO_4. **f**, The FE of CO_2R products (carbon monoxide, methane, formate, ethylene, ethanol (EtOH), *n*-propanol (PrOH) and acetate) and H_2 at different current densities in a Cu–Ag-based MEA equipped with ICM. The loading of Ag is 5 nm. For **e** and **f**, the plotted values are means of three replicates, while error bars indicate the standard deviation.

through the Donnan exclusion effect. This boosts local OH^- concentration, increasing the local pH at the catalyst surface and favouring CO_2R (Fig. 2c)[31,32]. High-resolution transmission electron microscopy (HRTEM) revealed the presence of a 5–7 nm homogeneous and conformal coating of ICM over the catalyst (Fig. 2d). Scanning transmission electron microscopy (STEM) with the elemental energy-dispersive X-ray spectroscopy (EDS) mapping confirmed a homogeneous distribution of potassium over the entire catalyst, establishing the ICM (Supplementary Fig. 7). Cations are expected to accumulate at the OHP with decreasing concentration through the ICM, which results in a cation accumulation distance that is much wider than the subnanometre distance estimated without an ICM (Fig. 2b).

Multi-carbon production at high energy and carbon efficiency

To probe the efficacy of the ICM in promoting C–C coupling, we studied CO_2R in an acid-fed MEA electrolyser. The Cu catalyst on GDE was equipped with an ICM and AEL. The AEL is sprayed on top of the ICM at a thickness of about 5 µm (Supplementary Fig. 8). We first optimized the cation availability on the catalyst surface by tuning the cation loading in the ICM (for example, at a cation:PFSA blend ratio of 2). Increasing the cation loading from 0 to 4 µl cm^{-2} decreased the H_2 FE from 40% to 10% while improving the ethylene FE from 15% to a peak of 35% (Fig. 2e and Supplementary Fig. 9). Previous studies attribute the enhanced activation of CO_2 to electrostatic interactions with adsorbate dipoles or changes in surface charge density[22,33,34]. High concentrations of available cations in the electrical double layer (enabled by the ICM) and high pH (enabled by the AEL) are expected to stabilize CO_2 intermediates on the catalyst surface (Supplementary Figs. 10 and 11) and decrease the thermodynamic energy barrier to multi-carbon production[22,33,34]. In contrast to H^+ ions, cations result in less osmotic drag and less water at the catalyst surface[21]. Assuming water to be the proton source for HER[33,35] and that HER is unaffected by cation identity[34,36], a lower amount of water present, owing to the hydrophobic nature of the ICM, could be another reason for the low H_2 FE observed. Reduced water transport could also contribute to less flooding in the catalyst layer pores, which decreases the thin film thickness through which CO_2 diffuses from the gas phase to the catalyst surface and thereby accelerates CO_2R[37–39]. ICM loadings higher than 4 µl cm^{-2} decreased ethylene formation and promoted HER. At increasing ICM loading, a higher cation concentration is expected to be distributed on the catalyst surface which can reduce CO_2 solubility (via the Sechenov effect)[40,41]. This results in a higher amount of solvated water, which can also decrease the CO_2 concentration on the cathode surface[40,41]. We also examined the effect of cation identity in the ICM. At the same applied current density (200 mA cm^{-2}) and cation concentration (4 µl cm^{-2}), we noticed that Na^+ caused the highest H_2 FE and the lowest ethylene FE (Supplementary Fig. 12), K^+ showed the opposite performance, and Cs^+ exhibited a similar performance to that of K^+. This behaviour can be explained by the solvation size and the hydration number of these cations increasing in the following order: $Cs^+ < K^+ < Na^+$ (ref. 21). Cs^+ and K^+ are expected to reach a higher concentration at the OHP (Fig. 2b). Consequently, Cs^+ and K^+ result in the highest electric field strength (Fig. 2a)[21] and CO_2R rate[22]. The ICM containing K^+ was employed for subsequent testing. The product distribution for the Cu catalyst with the ICM revealed that C_{2+} generation can reach 75% FE at 200 mA cm^{-2} (Fig. 2f).

We assessed the ICM-equipped Cu catalyst carbon efficiency performance by constraining CO_2 availability (Supplementary Fig. 13). By reducing the CO_2 flow rate from 5.0 to 0.8 standard cubic centimetres per minute (SCCM) while operating at 200 mA cm^{-2}, the carbon

https://doi.org/10.1038/s44160-024-00662-x

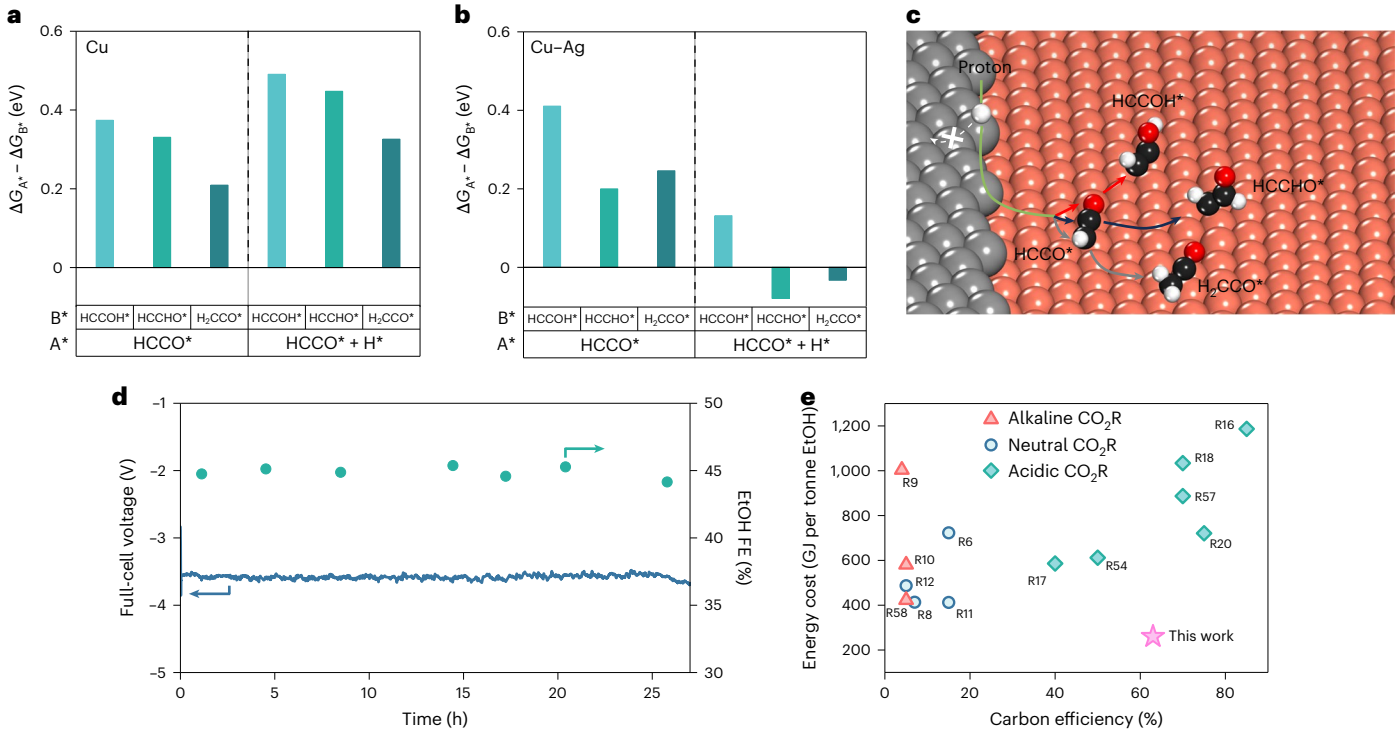


Fig. 4 | Mechanism of selective ethanol production and stable performance of ICM-equipped Cu–Ag catalyst in an MEA. a, The Gibbs free energy of HCCO* protonation on Cu via a PCET (left) or H* spillover mechanism (right). **b**, The Gibbs free energy of HCCO* protonation on Cu–Ag via a PCET (left) or H* spillover mechanism (right). **c**, Schematic illustrating the spillover mechanism of H* in a Cu–Ag catalyst, at the selectivity-determining step. Colour representation: Ag, grey; Cu, orange; C, black; O, red; H, white. **d**, CO_2R stability experiment for ethanol (EtOH) production with a Cu–Ag catalyst at 5 nm Ag loading and 200 mA cm^{-2}. The blue arrow corresponds to full-cell voltage and the teal arrow corresponds to the ethanol FE. **e**, A comparison of carbon efficiency and the ethanol production energy cost between the ICM-equipped Cu–Ag catalyst and the state-of-the-art CO_2R electrolysers used for ethanol production. The comparison is limited to reports implementing an MEA with a total current density of ≥100 mA cm^{-2} (refs. 6,8–12,16–18,20,54,57,58). R, reference.

efficiency increased from 11% to 61%. To enhance industrial viability of CO_2 electrolysers, reducing energy consumption is crucial. We then focused on improving energy efficiency at an industrially relevant current density (200 mA cm^{-2}). Using electrochemical impedance spectroscopy, we identified high ohmic losses, primarily from membrane and contact resistance. Thinner CEM membranes substantially reduced full-cell voltage, especially at higher current densities (Supplementary Note 3). By employing a catalyst-coated membrane, we achieved substantial voltage savings, crucial for commercial CO_2R feasibility. This approach was adopted for subsequent experiments. (Supplementary Note 3). The achieved $EE_{C_{2+}}$ (25%) represents a 2-fold increase compared with the best previous reports of MEA CO_2 electrolysers operating in a similarly high carbon efficiency regime (>60%) (Fig. 2g)[16,17,42]. The observed enhancement of C_{2+} production at high carbon efficiency in the case of the ICM-equipped Cu catalyst can be explained by sustained CO_2 availability (due to the hydrophobic nature of PFSA) and increased C–C coupling resulting from the local enrichment of the catalyst surface with alkali cations.

Bifurcation of the ethylene and ethanol pathways

To further enhance the selectivity of C_{2+} products, we sought to incorporate HER-suppressing Ag islands atop the Cu. We hypothesized that Ag clusters can act as CO-producing centres on the Cu surface while maintaining a low H* coverage, therefore limiting HER. A higher local CO production can increase the CO* coverage on Cu and facilitate the production of key C_2 intermediates (Fig. 3a)[43–47]. Using density functional theory (DFT) calculations, we found a decrease in the dimerization energy for C–C coupling when a higher concentration of CO exists on the Cu surface (Fig. 3b). To understand the influence of Ag on the coverage of H* on Cu surface, we theoretically investigated the hydrogen binding energy on Cu, Ag and Cu–Ag surface alloys of different composition and found that, upon incorporating Ag on Cu slabs, the binding energy and, by extension, the surface coverage of H* decrease substantially (Fig. 3c). The coverage of H* is directly related to HER activity through the rate-determining Volmer step, and its diminishing coverage can signal HER suppression. These findings suggested that decorating Ag on Cu surfaces tends to promote CO production and lower HER activity.Therefore, we experimentally designed a family of Cu–Ag catalysts in which pure Cu with a corrugated surface was sputtered with different loadings of Ag, yielding a surface with mixed Cu and Ag sites (Fig. 3d). STEM revealed the homogeneous distribution of Ag over the Cu surface. HRTEM images of Cu–Ag on PTFE further show uniformly decorated nanoclusters of Ag on the Cu surface, establishing extensive Cu–Ag interfaces (further characterizations are shown in Supplementary Figs. 14–21).

To assess the impact of the Ag loading, we evaluated the electrochemical performance of the ICM-equipped Cu–Ag catalyst at different Ag loadings in the acid-fed MEA electrolyser (Fig. 3e,f). Increasing the Ag loading on the Cu surface boosted ethanol production, with a concurrent decrease in ethylene formation. A Cu catalyst with 5-nm-thick Ag reached a peak ethanol FE of 45%, the highest achieved ethanol FE in acid-fed CO_2 electrolysers at an industrially relevant current density (200 mA cm^{-2}; Supplementary Table 1). We find that the previously well-understood pH independence of ethylene and ethanol formation no longer holds (Supplementary Fig. 22)[48–51]. This prompts the presence of a proton-transfer elementary step before the rate-determining step. We propose that the H* present at low coverages on Ag can transfer to the selectivity-determining C_2 intermediate on Cu at the Cu–Ag interface. As the H* formation on Ag is pH dependent, the change in ethylene versus ethanol selectivity becomes pH dependent. To probe

this hypothesis, we compare the H* spillover-assisted pathway on Cu and Cu–Ag at the selectivity-determining step.

It is accepted that CCO* is a key intermediate along the C_2 product pathways, and the proton-coupled electron transfer (PCET) on HCCO* can determine whether the ethanol or ethylene pathway will be followed[52]. Hence, we compare the hydrogenation of the oxygen, centre carbon and Cu-anchored carbon in HCCO* via direct PCET and a H* spillover mechanism on both Cu and Cu–Ag slabs (Fig. 4a,b and Supplementary Note 4)[43,53]. The protonation of the oxygen in HCCO* can lead to intermediates lying along both ethylene and ethanol pathways (HCCOH*), while the protonation of either the centre carbon resulting in HCHCO* or the Cu-anchored carbon resulting in H_2CCO* produces intermediates that lie exclusively along the ethanol pathway[52]. On Cu, the PCET pathway is more energetically favoured than the H* spillover pathway (Fig. 4a). The H* coverages on Cu are expected to be higher than those on Ag owing to the higher binding energy, and hence, the H* will facilitate the lower barrier HER reaction and not the energetically less favoured H* spillover. On the Cu–Ag system, the H* spillover pathway is more energetically accessible than the PCET pathway (Fig. 4b). Moreover, the spillover pathway renders the CHCHO* and H_2CCO* formation pathways exergonic, while the ethylene-producing HCCOH* remains endergonic. The result of this would be a shift in the selectivity-determining step towards ethanol formation, through a H* spillover mechanism (Fig. 4c). The Cu–Ag system is modelled with an Ag monolayer on a Cu surface with sufficient Cu empty sites for reaction intermediates, and the system closely mimics the Ag clusters present on the Cu–Ag catalyst. Higher loadings of Ag result in more Cu–$Ag_{cluster}$ boundaries, areas that were found to enable the H* spillover mechanism. The calculations also demonstrated that C–O bond scission is thermodynamically unfavourable on the interface of Cu–Ag clusters, which would favour ethanol production over ethylene (Supplementary Fig. 23). These findings further support the trend of ethylene and ethanol formation on Cu–Ag catalysts as a function of Ag loading (Fig. 3e), where the higher Ag loading resulted in more Cu–$Ag_{cluster}$ boundaries and higher ethanol FE, in parallel with ethylene suppression (Supplementary Fig. 24).

We also assessed the operating stability of the Cu–Ag catalyst equipped with an ICM in a f-BPM acidic MEA electrolyser (Fig. 4d). The electrolyser retained its initial ethanol FE (45%) after 26 h of continuous operation (Supplementary Fig. 25). By progressively reducing the flow rate of CO_2 from 5 to 0.8 SCCM, we achieved a carbon efficiency of about 63% towards CO_2R products (Supplementary Fig. 26). A carbonate-free electrolyser combined with the simultaneous achievement of high ethanol FE and high carbon efficiency resulted in a total production energy cost of 260 GJ per tonne ethanol. The ethanol production energy cost and carbon efficiency obtained through the ICM Cu–Ag approach outperform those of previous CO_2-to-ethanol reports (Fig. 4e and Supplementary Tables 1 and 2)[6,10–12,45,54]. As we move towards scaling up CO_2 electrolysers, we will indeed need to address challenges such as pressure differences, heterogeneity and heat management.

Conclusion

In this work, we present a catalyst microenvironment design principle for an acid-fed MEA electrolyser. Our findings demonstrate that enriching the catalyst surface with alkali cations while trapping locally generated OH^- using the ICM strategy enables the simultaneous achievement of high energy efficiency (25% towards C_{2+} products) and carbon efficiency (61%) in an acid-fed MEA electrolyser. By designing Cu–Ag catalysts and implementing the ICM strategy, we achieved electrosynthesis of ethanol in an MEA electrolyser with an FE of 45%, carbon efficiency of 63% and energy cost of 260 GJ per tonne ethanol simultaneously. This work presents CO_2 conversion to ethanol, with performance approaching industrial standards.

Methods

Chemicals and materials

Sulfuric acid (H_2SO_4; 98%), potassium sulfate (K_2SO_4; reagentPlus, ≥99.0%), potassium hydroxide (KOH; ACS Reagent, ≥98%, pellets), sodium hydroxide (NaOH; ACS Reagent, ≥85%, pellets), caesium hydroxide (CsOH; ACS Reagent, ≥98%, pellets), methanol, acetone and ethanol were purchased from Sigma-Aldrich. The anion-exchange ionomer powder (Aemion, AP2-CNN5-00-X) was purchased from Ionomr Innovations. All the Nafion membranes (117, 212 and XL) and PFSA dispersion (PFSA Dispersion (D5, 5%), alcohol based) were purchased from Fuel Cell Store. Titanium grade 2 felt, which was 60% porous before coating, was used to make the anode. The anode was coated with IrO_2 (Evoque, Magneto Special Anodes). Silver nanoparticles (Ag, 99.99%, 20 nm, metal basis) were acquired from US Research Nanomaterials. For the preparation of all electrolytes, deionized water (18.2 MΩ) was used. To sputter Cu catalyst onto the PTFE substrate (pore size of 450 nm, with a polypropylene support on the backside), a Cu target (>99.99%) was used in an Angstrom Nexdep sputtering system with a constant sputtering rate of 1.0 Å s^{-1} under vacuum (~10^{-6} Torr) until a catalyst thickness of 200 nm was achieved. The Cu–Ag catalysts were prepared by sputtering different Ag thicknesses on top of the Cu surface using an Ag target (>99.99%) with a constant sputtering rate of 0.1 Å s^{-1}. The nominal sputtering thickness of Ag was considered as Ag loading throughout the work.

GDE preparation

The GDEs (cathodes) equipped with an AEL were prepared by spraying a dilute Ameion solution to achieve the desired loading (3 mg cm^{-2}). The Ameion solution was prepared by dissolving 158 mg of the ionomer powder in 40 ml of a mixture of methanol and acetone (80 vol% + 20 vol%, respectively). This solution was diluted by doubling the total volume of the dispersion (with the same amount of methanol and acetone) before spraying and dried overnight before usage. To prepare cathodes equipped with ICM and the AEL, a desired amount of activated PFSA dispersion was spray coated on the cathode surface. The PFSA dispersion was prepared by diluting 40 µl of PFSA solution (5 wt%) in 2 ml methanol. To activate the PFSA with the cation, 80 µl of 1 M KOH, 1 M CsOH or 1 M NaOH was added to the PFSA dispersion and mixed for 10 min. This dispersion was stored overnight before the spray. After spray coating the ICM layer, the sample was allowed to air dry for at least 4 h before the AEL was sprayed.

The PFSA dispersion (5 wt% in solution) used in this work has the following properties: chemical formula $(C_7HF_{13}O_5S{\cdot}C_2F_4)_x$; molecular weight 554 g mol^{-1} and density 0.93 g ml^{-1}. The mass of PFSA on the cathode surface, given a PFSA loading of 2 µl cm^{-2}, can be estimated as

$$m_{\mathrm{PFSA}} = 0.002\,\frac{\mathrm{ml}}{\mathrm{cm}^2} \times 0.93\,\frac{\mathrm{g_s}}{\mathrm{ml}} \times \frac{5}{100}\,\frac{\mathrm{g_{PFSA}}}{\mathrm{g_s}} = 0.093 \times 10^{-3}\ \mathrm{mg\ cm^{-2}}.$$

The mass fraction of sulfonic acid groups (x_s) within the PFSA can be calculated as follows, considering 1 mol of PFSA as the reference and a molecular weight of 81 g mol^{-1} for a sulfonic group ($-SO_3H$):

$$x_{\mathrm{s}} = \frac{81\ \mathrm{g\,mol^{-1}}}{554\ \mathrm{g\,mol^{-1}}} = 0.15.$$

The number of moles of sulfonic acid groups is determined as

$$n_{\mathrm{s}} = \frac{9.3 \times 10^{-5}\ \mathrm{g\ cm^{-2}} \times 0.15}{81\ \mathrm{g\,mol^{-1}}} = 1.72 \times 10^{-7}\ \mathrm{mol\ cm^{-2}}.$$

This represents the lower limit of K^+ that is used to neutralize the sulfonic acid groups (1.72×10^{-7} M K^+ on the cathode surface). Considering 2 µl of PFSA solution, 4 µl of 1 M KOH was added to neutralize the sulfonic acid groups. Assuming all K^+ was spray coated on the Cu

surface, 4.0×10^{-6} mol cm^{-2} of K^+ was sprayed onto the catalyst surface, representing an excess amount used to stoichiometrically neutralize 1.72×10^{-7} mol cm^{-2} of sulfonic acid groups. Thus, the upper limit of K^+ concentration on the cathode surface can be considered as 4.0 mM.

Materials characterization

Scanning electron microscopy imaging and EDS X-ray mapping were performed via an Ultra-High-Resolution Schottky SEM SU7000, Hitachi microscope. A field emission transmission electron microscope (HF3300, Environmental-CFE-TEM) was used to capture the STEM, HRTEM images and EDS elemental mapping of Cu electrodes equipped with ICM and coated with Ag clusters. The crystalline structure of catalysts was examined with Rigaku Miniflex 600 6G Benchtop powder X-ray diffraction (step 0.01°, speed 1.5° min^{-1}). A Renishaw inVia Raman microscope equipped with a water immersion objective was employed for operando Raman spectroscopy. The spectra were obtained by employing a 785 nm laser, with a laser intensity set to 0.05% full laser intensity. The integration time was 0.1 s, and the measurements were repeated 100 times. For the measurements, an open-structured flow cell was utilized. As for the electrodes, an Ag/AgCl electrode filled with a saturated aqueous KCl solution served as the reference electrode, while a platinum wire was used as the counter electrode.

The in situ and ex situ X-ray absorption spectroscopy (XAS) measurements were conducted at the Hard X-ray Micro-Analysis Beamline at the Canadian Light Source. The in situ XAS spectra of the Cu K edge were recorded in fluorescence mode using a custom in situ XAS flow cell. The ex situ spectra were recorded in transmission mode. The energy calibration was performed using standard metallic Cu foil (8,979 eV). During measurements, 0.01 M H_2SO_4 + 0.1 M KCl aqueous solution was applied for the catholyte and 0.01 M H_2SO_4 for the anolyte. A Nafion 117 CEM separated the two electrolyte compartments. The measured catalysts included sputtered Cu (200 nm) with different loadings of silver on a PTFE substrate. A Pt anode and an Ag/AgCl (saturated KCl) electrode were used as the counter and reference electrodes, respectively. The electrolytes were flowed using peristaltic pumps at a flow rate of 1 ml min^{-1}. Pure CO_2 gas was supplied to the cathode during test.

MEA and CO_2 electrolysis

All CO_2R experiments were performed in a 1 cm^2 cell prepared with a grade 2 titanium cathode and anode flow fields. To construct the f-BPM, a commercial Nafion membrane was used as the CEM on top of the prepared AEL. IrO_2 felt was used as the anode. Electrochemical measurements were performed with an Autolab electrochemical workstation (PGSTAT204 equipped with a 10A booster). Throughout all experiments, unless otherwise specified, CO_2 was fed to the electrolyser at 5 SCCM using a mass flow controller. The anode side of the electrolyser was fed with 0.01 M H_2SO_4 for all experiments, unless otherwise specified. The cell voltages reported in this work are full-cell voltages without *iR* correction (*i*, current; *R*, resistance). For the flow-cell setup, the Cu–Ag catalyst GDE was used as the working electrode (cathode), an IrO_2 felt as the counter electrode (anode) and an Ag/AgCl electrode as the reference electrode. A CEM (Nafion 117) was used to separate the anodic and cathodic compartments. The catholyte and anolyte were 0.01 M H_2SO_4 + 3 M KCl and 0.01 M H_2SO_4, respectively. The stability experiment involved gradually increasing the applied current, starting at 100 mA cm^{-2} for an initial 10 min, followed by a transition to 200 mA cm^{-2} for the remaining duration.

CO_2R product analysis

The gas outlet stream (containing CO_2R products and unreacted CO_2) was sampled for product analysis. A gas chromatograph (PerkinElmer Clarus 590) coupled with a thermal conductivity detector and flame ionization detector was utilized for this purpose. The chromatograph contains a Molecular Sieve 5A Capillary Column and a packed Carboxen-1000 Column, with argon serving as the carrier gas. The flow rates of gas in and out of the cell were determined via a bubble column. The FE of gas products was determined using

$$\mathrm{FE_{gas}}\,(\%) = \frac{z \times r \times F \times v}{I \times V_\mathrm{m}},$$

where z corresponds to the number of electrons involved in the production of the gas product in question, r represents the concentration of detected gas in ppm, F is the Faraday's constant (96,485 C mol^{-1}), v is the gas flow rate, I represents the total current and V_m is the unit molar volume of gas.

Proton nuclear magnetic resonance spectroscopy was utilized to measure the quantities of liquid products. This was performed on an Agilent DD2 500 spectrometer while in D_2O water suppression mode. Dimethyl sulfoxide was employed as an internal standard. The FE of liquid products was determined using

$$\mathrm{FE_{liquid}}\,(\%) = \frac{z \times F \times n_\mathrm{product}}{I \times t},$$

where n_product is the total quantity of products measured in moles, and $I \times t$ (C) denotes the total charge (Q) that was passed during the experiment.

Carbon and energy efficiency calculation

Carbon efficiency (or SPCE; the percentage of CO_2 converted per total CO_2 input), which pertains to gas, liquid or a combination of gas and liquid products at 25 °C and 1 atm, was determined via

$$\text{Carbon efficiency}\,(\%) = \frac{(j_\mathrm{product} \times 60\ \mathrm{s})/(n \times F)}{(v \times 1\ \mathrm{min})/V_\mathrm{m}},$$

where j_product is the partial current density of a specific group of CO_2R products (measured in amperes), v is the gas flow rate (l min^{-1}) and V_m is equivalent to 24.05 (l mol^{-1}). The full-cell EE for each product was determined using

$$\mathrm{EE_{product}} = \frac{\left(1.23\ \mathrm{V} + \left(-E^0_\mathrm{product}\right)\right) \times \mathrm{FE_{product}}}{-E_\mathrm{cell}}.$$

Here, E^0_product refers to the thermodynamic potential for the formation of the specific CO_2R product, and E_cell is the full-cell voltage obtained from the MEA electrolyser without *iR* correction.

Multiphysics modelling

The local concentrations of cations at different current densities, the local pH at the cathode surface with different configurations, and the electric field at the electrode surface were modelled in COMSOL (COMSOL Multiphysics version 5.5). A simulation tutorial was provided in our previous report[55]. See Supplementary Information for more details.

DFT calculation

DFT was applied to calculate the Gibbs free energy diagrams of CO_2R and HER on the (111) surfaces of Cu (Cu(111)) and Ag clusters on Cu surface (Ag–Cu(111)) as well as the charge transfer between the HCCOH* intermediate and surface. All the DFT calculations were carried out in the Vienna ab initio simulation package with a plane wave pseudo-potential implementation. See Supplementary Information for more details.

Data availability

All data supporting the findings of this study, including computational specifics, and other experimental and microscopic analyses, can be found in the Article and its Supplementary Information.

References

1. Masel, R. I. et al. An industrial perspective on catalysts for low-temperature CO_2 electrolysis. *Nat. Nanotechnol.* **16**, 118–128 (2021).
2. Gao, D., Arán-Ais, R. M., Jeon, H. S. & Roldan Cuenya, B. Rational catalyst and electrolyte design for CO_2 electroreduction towards multicarbon products. *Nat. Catal.* **2**, 198–210 (2019).
3. Nguyen, T. N. et al. Catalyst regeneration via chemical oxidation enables long-term electrochemical carbon dioxide reduction. *J. Am. Chem. Soc.* **144**, 13254–13265 (2022).
4. Zhang, S., Fan, Q., Xia, R. & Meyer, T. J. CO_2 reduction: from homogeneous to heterogeneous electrocatalysis. *Acc. Chem. Res.* **53**, 255–264 (2020).
5. Ethanol Market - Global Industry Analysis, Size, Share, Growth, Trends and Forecast 2017–2022. *Transparency Market Research* (2018). https://www.transparencymarketresearch.com/ethanol-market.html.
6. Gabardo, C. M. et al. Continuous carbon dioxide electroreduction to concentrated multi-carbon products using a membrane electrode assembly. *Joule* **3**, 2777–2791 (2019).
7. Wang, P. et al. Boosting electrocatalytic CO_2-to-ethanol production via asymmetric C–C coupling. *Nat. Commun.* **13**, 1–11 (2022).
8. Gu, Z. et al. Efficient electrocatalytic CO_2 reduction to C_{2+} alcohols at defect-site-rich Cu surface. *Joule* **5**, 429–440 (2021).
9. Adnan, M. A. et al. Directly-deposited ultrathin solid polymer electrolyte for enhanced CO_2 electrolysis. *Adv. Energy Mater.* https://doi.org/10.1002/aenm.202203158 (2023).
10. Shang, L. et al. Efficient CO_2 electroreduction to ethanol by Cu_3Sn catalyst. *Small Methods* **6**, 2101334 (2022).
11. Li, F. et al. Cooperative CO_2-to-ethanol conversion via enriched intermediates at molecule–metal catalyst interfaces. *Nat. Catal.* **3**, 75–82 (2019).
12. Wang, X. et al. Efficient electrically powered CO_2-to-ethanol via suppression of deoxygenation. *Nat. Energy* **5**, 478–486 (2020).
13. Kim, C. et al. Tailored catalyst microenvironments for CO_2 electroreduction to multicarbon products on copper using bilayer ionomer coatings. *Nat. Energy* **6**, 1026–1034 (2021).
14. Rabinowitz, J. A. & Kanan, M. W. The future of low-temperature carbon dioxide electrolysis depends on solving one basic problem. *Nat. Commun.* **11**, 10–12 (2020).
15. Ma, M. et al. Insights into the carbon balance for CO_2 electroreduction on Cu using gas diffusion electrode reactor designs. *Energy Environ. Sci.* **13**, 977–985 (2020).
16. O'Brien, C. P. et al. Single pass CO_2 conversion exceeding 85% in the electrosynthesis of multicarbon products via local CO_2 regeneration. *ACS Energy Lett.* **6**, 2952–2959 (2021).
17. Xu, Y. et al. A microchanneled solid electrolyte for carbon-efficient CO_2 electrolysis. *Joule* **6**, 1333–1343 (2022).
18. Huang, J. E. et al. CO_2 electrolysis to multicarbon products in strong acid. *Science* **372**, 1074–1078 (2021).
19. Fan, M. et al. Cationic-group-functionalized electrocatalysts enable stable acidic CO_2 electrolysis. *Nat. Catal.* **6**, 763–772 (2023).
20. Zhao, Y. et al. Conversion of CO_2 to multicarbon products in strong acid by controlling the catalyst microenvironment. *Nat. Synth.* **2**, 403–412 (2023).
21. Bohra, D., Chaudhry, J. H., Burdyny, T., Pidko, E. A. & Smith, W. A. Modeling the electrical double layer to understand the reaction environment in a CO_2 electrocatalytic system. *Energy Environ. Sci.* **12**, 3380–3389 (2019).
22. Liu, M. et al. Enhanced electrocatalytic CO_2 reduction via field-induced reagent concentration. *Nature* **537**, 382–386 (2016).
23. Yang, B. et al. Accelerating CO_2 electroreduction to multicarbon products via synergistic electric–thermal field on copper nanoneedles. *J. Am. Chem. Soc.* **144**, 3039–3049 (2022).
24. Yang, K. et al. Cation-driven increases of CO_2 utilization in a bipolar membrane electrode assembly for CO_2 electrolysis. *ACS Energy Lett.* **6**, 4291–4298 (2021).
25. Pan, B. et al. Close to 90% single-pass conversion efficiency for CO_2 electroreduction in an acid-fed membrane electrode assembly. *ACS Energy Lett.* **7**, 4224–4231 (2022).
26. Garg, S. et al. How alkali cations affect salt precipitation and CO_2 electrolysis performance in membrane electrode assembly electrolyzers. *Energy Environ. Sci.* **16**, 1631–1643 (2023).
27. Konwar, L. J., Mäki-Arvela, P. & Mikkola, J. P. SO_3H-containing functional carbon materials: synthesis, structure, and acid catalysis. *Chem. Rev.* **119**, 11576–11630 (2019).
28. Kusoglu, A. & Weber, A. Z. New insights into perfluorinated sulfonic-acid ionomers. *Chem. Rev.* **117**, 987–1104 (2017).
29. Shi, S., Liu, Z., Lin, Q., Chen, X. & Kusoglu, A. Role of ionic interactions in the deformation and fracture behavior of perfluorosulfonic-acid membranes. *Soft Matter* **16**, 1653–1667 (2020).
30. Shi, S., Weber, A. Z. & Kusoglu, A. Structure–transport relationship of perfluorosulfonic-acid membranes in different cationic forms. *Electrochim. Acta* **220**, 517–528 (2016).
31. Jervis, R. et al. The importance of using alkaline ionomer binders for screening electrocatalysts in alkaline electrolyte. *J. Electrochem. Soc.* **164**, F1551–F1555 (2017).
32. Lees, E. W. et al. Linking gas diffusion electrode composition to CO_2 reduction in a flow cell. *J. Mater. Chem. A* **8**, 19493–19501 (2020).
33. Monteiro, M. C. O. et al. Absence of CO_2 electroreduction on copper, gold and silver electrodes without metal cations in solution. *Nat. Catal.* **4**, 654–662 (2021).
34. Ringe, S. et al. Understanding cation effects in electrochemical CO_2 reduction. *Energy Environ. Sci.* **12**, 3001–3014 (2019).
35. Ooka, H., Figueiredo, M. C. & Koper, M. T. M. Competition between hydrogen evolution and carbon dioxide reduction on copper electrodes in mildly acidic media. *Langmuir* **33**, 9307–9313 (2017).
36. Resasco, J. et al. Promoter effects of alkali metal cations on the electrochemical reduction of carbon dioxide. *J. Am. Chem. Soc.* **139**, 11277–11287 (2017).
37. Weng, L.-C., Bell, A. T. & Weber, A. Z. Modeling gas-diffusion electrodes for CO_2 reduction. *Phys. Chem. Chem. Phys.* **20**, 16973–16984 (2018).
38. Li, G. et al. Electrode engineering for electrochemical CO_2 reduction. *Energy Fuels* **36**, 4234–4249 (2022).
39. Li, M. et al. The role of electrode wettability in electrochemical reduction of carbon dioxide. *J. Mater. Chem. A* **9**, 19369–19409 (2021).
40. Ozden, A. et al. Energy- and carbon-efficient CO_2/CO electrolysis to multicarbon products via asymmetric ion migration–adsorption. *Nat. Energy* **8**, 179–190 (2023).
41. García de Arquer, F. P. et al. CO_2 electrolysis to multicarbon products at activities greater than 1 A cm^{-2}. *Science* **367**, 661–666 (2020).
42. Xie, K. et al. Bipolar membrane electrolyzers enable high single-pass CO_2 electroreduction to multicarbon products. *Nat. Commun.* **13**, 3609 (2022).
43. Zheng, Y. et al. Understanding the roadmap for electrochemical reduction of CO_2 to multi-carbon oxygenates and hydrocarbons on copper-based catalysts. *J. Am. Chem. Soc.* **141**, 7646–7659 (2019).
44. Ma, W. et al. Electrocatalytic reduction of CO_2 to ethylene and ethanol through hydrogen-assisted C–C coupling over fluorine-modified copper. *Nat. Catal.* **3**, 478–487 (2020).

45. Li, Y. C. et al. Binding site diversity promotes CO_2 electroreduction to ethanol. *J. Am. Chem. Soc.* **141**, 8584–8591 (2019).
46. Clark, E. L., Hahn, C., Jaramillo, T. F. & Bell, A. T. Electrochemical CO_2 reduction over compressively strained CuAg surface alloys with enhanced multi-carbon oxygenate selectivity. *J. Am. Chem. Soc.* **139**, 15848–15857 (2017).
47. Morales-Guio, C. G. et al. Improved CO_2 reduction activity towards C_{2+} alcohols on a tandem gold on copper electrocatalyst. *Nat. Catal.* **1**, 764–771 (2018).
48. Chang, X., Zhao, Y. & Xu, B. pH dependence of Cu surface speciation in the electrochemical CO reduction reaction. *ACS Catal.* **10**, 13737–13747 (2020).
49. Hou, J., Chang, X., Li, J., Xu, B. & Lu, Q. Correlating CO coverage and co electroreduction on cu via high-pressure in situ spectroscopic and reactivity investigations. *J. Am. Chem. Soc.* **144**, 22202–22211 (2022).
50. Chang, X. et al. C–C coupling is unlikely to be the rate-determining step in the formation of C_{2+} products in the copper-catalyzed electrochemical reduction of CO. *Angew. Chem. Int. Ed.* **61**, e202111167 (2022).
51. Li, J. et al. Electrokinetic and in situ spectroscopic investigations of CO electrochemical reduction on copper. *Nat. Commun.* **12**, 1–11 (2021).
52. Santatiwongchai, J., Faungnawakij, K. & Hirunsit, P. Comprehensive mechanism of CO_2 electroreduction toward ethylene and ethanol: the solvent effect from explicit water–Cu(100) interface models. *ACS Catal.* **11**, 9688–9701 (2021).
53. Nitopi, S. et al. Progress and perspectives of electrochemical CO_2 reduction on copper in aqueous electrolyte. *Chem. Rev.* **119**, 7610–7672 (2019).
54. Miao, R. K. et al. Electroosmotic flow steers neutral products and enables concentrated ethanol electroproduction from CO_2. *Joule* **5**, 2742–2753 (2021).
55. Li, F., Zhou, C. & Klinkova, A. Simulating electric field and current density in nanostructured electrocatalysts. *Phys. Chem. Chem. Phy.* https://doi.org/10.1039/d2cp02846h (2022).
56. Lark, T. J. et al. Environmental outcomes of the US Renewable Fuel Standard. *Proc. Natl Acad. Sci. USA* **119**, e2101084119 (2022).
57. Xie, Y. et al. High carbon utilization in CO_2 reduction to multi-carbon products in acidic media. *Nat. Catal.* **5**, 564–570 (2022).
58. Wang, X. et al. Ag^+-doped InSe nanosheets for membrane electrode assembly electrolyzer toward large-current electroreduction of CO_2 to ethanol. *Angew. Chem. Int. Ed.* **62**, e202313646 (2023).

Acknowledgements

We acknowledge funding for this work from the Government of Canada's New Frontiers in Research Fund (NFRF), CANSTOREnergy project NFRFT-2022-00197, NRC and the University of Toronto Collaboration Centre Program in Green Energy Materials (CC-GEM, GEM-PRJ-01), Ontario Research Foundation Research Excellence Program and the CIFAR Bio-Inspired Solar Energy programme. D.S. acknowledges the support received from the Canada Research Chairs programme. A.S.Z. thanks NSERC for a Postdoctoral Fellowship. T.A. acknowledges graduate scholarships and fellowships from Hatch, NSERC and the University of Toronto. We also acknowledge Compute Canada for providing computing resources. We appreciate the Ontario Centre for the Characterization of Advanced Materials (OCCAM) at the University of Toronto for conducting microscopy characterizations. We thank M. Chehelamirani for his assistance in preparing the graphical abstract.

Author contributions

D.S. and E.H.S. supervised the project. A.S.Z. conceived the idea and performed the electrochemical experiments and analyses. A.S.Z. and T.A. co-wrote the manuscript. F.L. performed the DFT calculations. F.L. and T.A. carried out the FEM simulation. R.D. assisted with the DFT calculation and manuscript writing. E.S., F.A., C.P.O., R.K.M. and D.Y. helped with electrochemical experiments. J.K. assisted with TEM imaging. A.O. and Y.Z. assisted with the catalyst preparation. P.P., D.K. and S.P. carried out the XAS measurements. M.Z. assisted with the schematics. L.F. carried out the X-ray diffraction measurement. C.M.G., J.P.E. and A.H.I. assisted with the research advice, discussions and editing. All authors contributed to the manuscript editing.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s44160-024-00662-x.

Correspondence and requests for materials should be addressed to David Sinton.

Peer review information *Nature Synthesis* thanks Remco Hartkamp, Yanguang Li and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Alexandra Groves, in collaboration with the *Nature Synthesis* team.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2024

nature synthesis

Supplementary information https://doi.org/10.1038/s44160-024-00662-x

Carbon- and energy-efficient ethanol electrosynthesis via interfacial cation enrichment

In the format provided by the authors and unedited

Table of Contents

1. Supplementary Methods

1.1 Multiphysics simulations 1

1.2 Density functional theory (DFT) calculations 7

2. Supplementary Notes

2.1 Supplementary Note 1 | Different MEA electrolyzer design and local microenvironment 9

2.2 Supplementary Note 2 | Discussion on Surface-Enhanced Raman Spectroscopy – cation effect 14

2.3 Supplementary Note 3 | Energy efficiency/full-cell voltage reduction 24

2.4 Supplementary Note 4 | DFT modelling of Cu-Ag interface for CO_2R 35

3. Supplementary Figures

Fig. 1 | Different configurations of CO_2 MEA electrolyzers 10

Fig. 2 | CO_2R performance in MEA electrolyzers at different anolyte concentration 11

Fig. 3 | Modelling results of local pH in the presence of anion exchange layer 12

Fig. 4 | Effect of cation concentration on the electric field at the cathode surface 13

Fig. 5 | *Operando* Raman analysis of Cu catalyst in acidic electrolyte 15

Fig. 6 | Operando Raman analysis – effect of cation 16

Fig. 7 | SEM characterization of Cu catalyst coated with ICM 17

Fig. 8 | SEM of anion exchange layer 18

Fig. 9 | Effect of ICM loading on CO_2R performance 19

Fig. 10 | Schematic of electrolyzer design and EDL for f-BPM MEA electrolyzer equipped with ICM 20

Fig. 11 | *Operando* Raman analysis of Cu catalyst equipped with ICM 21

Fig. 12 | Effect of alkali cation type in the ICM on the CO_2R performance 22

Fig. 13 | Carbon efficiency of Cu catalyst equipped with ICM 23

Fig. 14 | SEM characterization of Cu catalyst 26

Fig. 15 | SEM characterization of Cu- 2 nm Ag catalyst 27

Fig. 16 | SEM characterization of Cu- 5 nm Ag catalyst 28

Fig. 17 | SEM characterization of Cu- 10 nm Ag catalyst 29

Fig. 18 | STEM and HRTEM images of Cu- 5 nm Ag catalyst at different magnifications 30

Fig. 19 | XRD patterns of catalysts 31

Fig. 20 | High resolution XPS spectra 32

Fig. 21 | Ex-situ and in-situ X-ray absorption spectroscopy (XAS) spectra 33

Fig. 22 | Effect of pH on ethanol production 34

Fig. 23 | Electron density difference plots for Cu and Cu-Ag catalysts 36
Fig. 24 | Effect of deposition of AgNP on CO_2R performance 37
Fig. 25 | Stability of Cu-5Ag catalyst with and without the ICM adlayer 38
Fig. 26 | Carbon efficiency of a Cu-Ag catalyst 39
Fig. 27 | Multiphysics simulation domain 40
Fig. 28 | Full-cell voltage for acid-fed CO_2 MEA electrolyzer with different CEM membranes 41

4. Supplementary Tables

Table 1 | Total energy cost comparison for ethanol production 42
Table 2 | Ethanol FE in recent literature of CO_2R 43
Table 3 | Diffusion-coefficients for modelling 44
Table 4 | Solvation sizes 45
Table 5 | Rate constants used for modelling 46
Table 6 | h parameters used to estimate Sechenov's constant 47
Table 7 | Contributions to Gibbs free energies of gas species and adsorbates 48
Table 8 | Full cell voltage and RI for MEA electrolyzer 49

1. Supplementary Methods

1.1 Multiphysics simulations

The simulation of cation distributions was performed in the COMSOL Multiphysics package based on the finite-element-method solver. The electric double layer at the electrode-electrolyte interface was modelled using the Gouy-Chapman-Stern (GCS) model, [1] which consists of the Stern layer, the diffuse layer, and diffusion layer (**Supplementary Figure 27**). The Stern layer consists of a monolayer of surface-adsorbed hydrated cations on the electrode surface, with a thickness of d_s depending on the hydrated cation, which is set to be 0.28 nm for H^+, 0.358 nm for Na^+, 0.33 nm for K^+, and 0.329 nm for Cs^+. The diffuse layer and diffusion layers contain free anions and cations distributed according to the Poisson and Boltzmann law, which forms the concentration gradient away from the electrode surface. In addition, the ICM and AEM were also considered in this work, specifically for pH simulations.

Generalised modified Poisson-Nernst-Plank (GMPNP)[2–4] with the homogeneous reaction term were applied to calculate the cations distribution, E-field, and pH in the diffuse and diffusion layer. The GMPNP equations are given by:

$$\frac{\partial C_i}{\partial t} = -\nabla \cdot \boldsymbol{J_i} + \sum_p R_i \qquad \textbf{(Equation S1)}$$

$$\boldsymbol{J_i} = D_i \nabla C_i - \frac{D_i C_i z_i F}{RT} \nabla \phi - \frac{D_i C_i N_A a_i^3 \sum_i \nabla C_i}{1 - \sum_i a_i^3 C_i} \qquad \textbf{(Equation S2)}$$

$$\nabla \cdot (\varepsilon_0 \varepsilon_e \nabla \phi) = -\rho_f = -F \sum_i z_i C_i \qquad \textbf{(Equation S3)}$$

In Equation S1, C_i is the concentration of the ith species, $\boldsymbol{J_i}$ is the flux for the ith species, and R_i is the rate of production of the ith species due to the homogeneous reaction as given by equation (S9)-(S13). In Equation S2, D_i is the diffusion constant of the ith species, a_i is the

solvation size of the ith species, z_i is the valence charge of the ith species, F is the Faradaic constant, R is the ideal gas constant, T is the temperature, N_A is the Avogadro constant, and ϕ is the space potential in the electrolyte. In Equation S3, ε_0 is the vacuum permittivity, and ε_e is the relative permittivity of the electrolyte solution.

As defined in Gouy-Chapman-Stern (GCS) model,[5] the OHP is the closet proximity of the ions to the electrode surface. Therefore, the Stern layer contains no charged species but water. Thus, the charge density at OHP, σ_{OHP}, is given by:[3]

$$\sigma_{OHP} = \varepsilon_0 \varepsilon_e \left.\frac{d\phi}{dx}\right|_{x_{OHP}} = \frac{\varepsilon_0 \varepsilon_{Stern}(\phi_M - \phi_{OHP})}{d_{Stern}} \qquad \textbf{(Equation S4)}$$

where ε_{Stern} and d_{Stern} denote the relative permittivity in and thickness of the Stern layer, respectively, ϕ_M and ϕ_{OHP} are the potentials at the electrode surface and OHP, respectively. Equation S4 specifies the relationship between the potential (ϕ) and the potential gradient ($\frac{d\phi}{dx}$) at the OHP.

To account for the E-field dependence of the dielectric permittivity in the diffuse and diffusion layers, the Booth model, equation (S5), is employed. [6–8]

$$\varepsilon_e(E;\beta) = n^2 + (\varepsilon_r(0) - n^2)\frac{3}{\beta E}\left[\coth(\beta E) - \frac{1}{\beta E}\right] \qquad \textbf{(Equation S5)}$$

In equation S5, $E = |-\nabla\phi(r)|$ is the local E-field strength, n is the electrolyte refractive index, and β is a parameter given by: [9]

$$\beta = \frac{5\mu}{2kT}(n^2 + 2) \qquad \textbf{(Equation S6)}$$

where μ is the dipole moment of the water molecule.

Furthermore, the value of ε_{Stern} was extracted from Equation S7 and S8 as suggested by Johnson et al.[3] to consider the effect of the electric field on dielectric permittivity in the stern layer.

$$C_{Stern} = \frac{\varepsilon_0 \varepsilon_{Stern}}{x_{Stern}} \qquad \textbf{(Equation S7)}$$

$$C_{Stern} = -1.8923\, V^3 - 4.4698\, V^2 + 2.0741V + 26.323 \qquad \textbf{(Equation S8)}$$

Even though Equation S8 being derived from experiments in Na electrolyte, it provides a more plausible estimation of ε_{Stern} when juxtaposed with the scenario of presuming ε_{Stern} equals ε_e.

The diffusion rate of each species within the ion-exchange layer is reduced compared to that in the solution. Given its porous structure, the effective diffusivity should be accounted for within the porous domain during simulation, which depends on the porosity and tortuosity according to Bruggeman relationship.[10] More importantly, the fixed charges within the ion-exchange layer introduces a decelerating effect on the diffusion rate of the counter ionic species. Nevertheless, the precise quantitative assessment of the porosity, tortuosity, and fixed-charge density in the ion-exchange layer poses significant challenges. Consequently, we opted for a simplified modeling approach, presuming a tenfold reduction in the diffusion rate of counter ions within the ion-exchange layer. This choice provides a qualitative exploration of the impact of ion-exchange layer thickness on local pH variations using GMPNP model.

In the simulations, the reactions S9-S13 are considered for the homogeneous reactions occurring within the electrolyte solution.

$$CO_2 + H_2O \xleftrightarrow{k_{1f},\ k_{1r}} HCO_3^- + H^+ \quad (K_1) \qquad \textbf{(Equation S9)}$$

$$HCO_3^- \xleftrightarrow{k_{2f},\ k_{2r}} CO_3^{2-} + H^+ \quad (K_2) \qquad \textbf{(Equation S10)}$$

$$CO_2 + OH^- \overset{k_{3f},\ k_{3r}}{\longleftrightarrow} HCO_3^- \quad (K_3)$$ **(Equation S11)**

$$HCO_3^- + OH^- \overset{k_{4f},\ k_{4r}}{\longleftrightarrow} CO_3^{2-} + H_2O \quad (K_4)$$ **(Equation S12)**

$$H_2O \overset{k_{wf},\ k_{wr}}{\longleftrightarrow} H^+ + OH^- \quad (K_w)$$ **(Equation S13)**

The heterogeneous reactions S14 and S15 are considered as the electrochemical reactions occurring on the planar electrode surface during the CO_2RR process:

$$CO_2(aq) + H_2O + 2e^- \rightleftarrows \mathrm{CO(g)} + 2OH^-$$ **(Equation S14)**

$$2H_2O + 2e^- \rightleftarrows H_2(\mathrm{g}) + 2OH^-$$ **(Equation S15)**

The flux of the solution species at the surface is then given by equations (S16)-(S18):

$$\vec{J}_i\big|_{x=0,t} = 0 \quad i = HCO_3^-, CO_3^{2-}, K^+, H^+, Na^+, Cs^+$$ (**Equation S16**)

$$\vec{J}_{CO_2}\Big|_{x=0,t} = \frac{1}{n_{e,CO}} \times \frac{j_{tot}}{F} \times FE_{CO}$$ (**Equation S17**)

$$\vec{J}_{OH^-}\big|_{x=0,t} = \frac{1}{n_{e,OH^-}} \times \frac{j_{tot}}{F}$$ (**Equation S18**)

where F is the Faraday's constant, t is the time, $n_{e,CO}$ is the number of electrons consumed to reduce CO_2 to CO and n_{e,OH^-} is the number of electrons consumed per OH^-.

The rate of each species R_i, can be broken into carbonate equilibria as follows:

$$R_{CO_2} = \left(-[CO_2]_{aq}\,[H_2O]k_{1f} + [H^+][HCO3^-]\,k_{1r}\right) + \left(-[CO_2]_{aq}\,[OH^-]k_{3f} + [HCO3^-]\,k_{3r}\right)$$ **(Equation S19)**

$$R_{CO_3{}^{2-}} = \left([HCO_3{}^-]\,k_{2f} - [H^+]\left[CO_3{}^{2-}\right]k_{2r}\right) + \left([HCO_3{}^-]\,[OH^-]k_{4f} - [H_2O]\left[CO_3{}^{2-}\right]k_{4r}\right)$$ **(Equation S20)**

$$R_{HCO_3{}^-} = \left([CO_2]_{aq}\,[H_2O]k_{1f} - [H^+][HCO_3{}^-]\,k_{1r}\right) + \left(-[HCO_3{}^-]\,k_{2f} + [H^+]\left[CO_3{}^{2-}\right]k_{2r}\right) + \left([CO_2]_{aq}\,[OH^-]k_{3f} - [HCO_3{}^-]\,k_{3r}\right) + \left(-[HCO_3{}^-]\,[OH^-]k_{4f} + [H_2O]\left[CO_3{}^{2-}\right]k_{4r}\right)$$ **(Equation S21)**

$$R_{H^+} = \left([CO_2]_{aq}\,[H_2O]k_{1f} - [H^+][HCO_3{}^-]\,k_{1r}\right) + \left([HCO_3{}^-]\,k_{2f} - [H^+]\left[CO_3{}^{2-}\right]k_{2r}\right) + \left([H_2O]k_{wf} - [H^+][OH^-]\,k_{wr}\right)$$ **(Equation S22)**

$$R_{OH^-} = \left(-[CO_2]_{aq}\,[OH^-]k_{3f} + [HCO_3{}^-]\,k_{3r}\right) + \left(-[HCO_3{}^-]\,[OH^-]k_{4f} + [H_2O]\left[CO_3{}^{2-}\right]k_{4r}\right) + \left([H_2O]k_{wf} - [H^+][OH^-]\,k_{wr}\right)$$ **(Equation S23)**

The values of the rate constants for reactions S9-S13 are given in **Supplementary Table 5**. The gas concentration of CO_2 is assumed to obey Henry's law as given by

$$C^0_{CO_2,aq} = K^0_H\, C_{CO_2,g}$$ (**Equation S24**)

Where K^0_H is Henry's constant and is given as a function of temperature T by equation (S25):

$$\ln K^0_H = 93.2417 * \left(\frac{100}{T}\right) - 60.2409 + 23.3585 * \ln\left(\frac{T}{100}\right)$$ (**Equation S25**)

where T is the temperature and is assumed to be 298.15 K in this study. The saturated concentration of CO_2 in an electrolyte is then given by equation (S26):

$$\log\left(\frac{C^0_{CO_2,aq}}{C_{CO_2,aq}}\right) = K_S C_S$$ (**Equation S26**)

In equation (S26), C_S is the molar concentration of the electrolyte and K_S is the Sechenov's constant and can be calculated by the Equation S27:

$$K_S = \sum(h_{CO_2} + h_{ion})$$ (**Equation S27**)

$$h_{CO_2} = h^0_{CO_2} + h^T_{CO_2}(T - 298.15)$$ (**Equation S28**)

Values of h for all species can be found in **Supplementary Table 6**.

The intricacy of the EDL is heightened by factors such as Faradic currents, interfacial chemical interactions, and phenomena at molecular scales, where quantum mechanics becomes more pertinent than continuum theory. Additionally, the distinct behavior of hydrated ions under

a strong electric field, as compared to in a dilute bulk solution, also complicates the study of the EDL, especially with adsorption. Existing models like MPB, GMPNP, and SMPNP, while well-established, do not completely encapsulate the complexity of the EDL. Nonetheless, these models remain useful tools for visualization and modeling of the EDL. The GMPNP and SMPNP models offer more reliable and flexible approaches for modeling the EDL compared to the MPB model. In light of this, the GMPNP model was implemented to describe the EDL in this paper, reflecting a more accurate representation of EDL phenomena.

1.2 Density functional theory (DFT) calculations. DFT was applied to calculate the Gibbs free energy change of various HCCO* protonation processes, including HCCO* to HCCOH*, HCCO* to HCCHO*, HCCO* to H_2CCO*, on Cu and Cu-Ag surfaces. The structure of HCCO* + H* was also calculated to study the effect of adsorbed H* on the HCCO* protonation processes. All the DFT calculations were carried out in the Vienna ab initio simulation package[11–13] with a plane wave pseudo-potential implementation[14]. The exchange-correlation function was described by the spin-polarized generalized gradient approximation of Perdew-Burker-Ernzerhof[14] and the electron-ion interactions was described by projector augmented wave potentials[15,16]. The kinetic cut-off energy of 450 eV was used for the plane-wave expansion. The $6 \times 3 \times 1$ and $12 \times 6 \times 1$ Monkhorst-Pack[17] k-point meshes were used for geometry optimization and electronic structure calculation. The long-range Van der Waals interactions were described by the zero-damping DFT-D3 method of Grimme et al.[18]

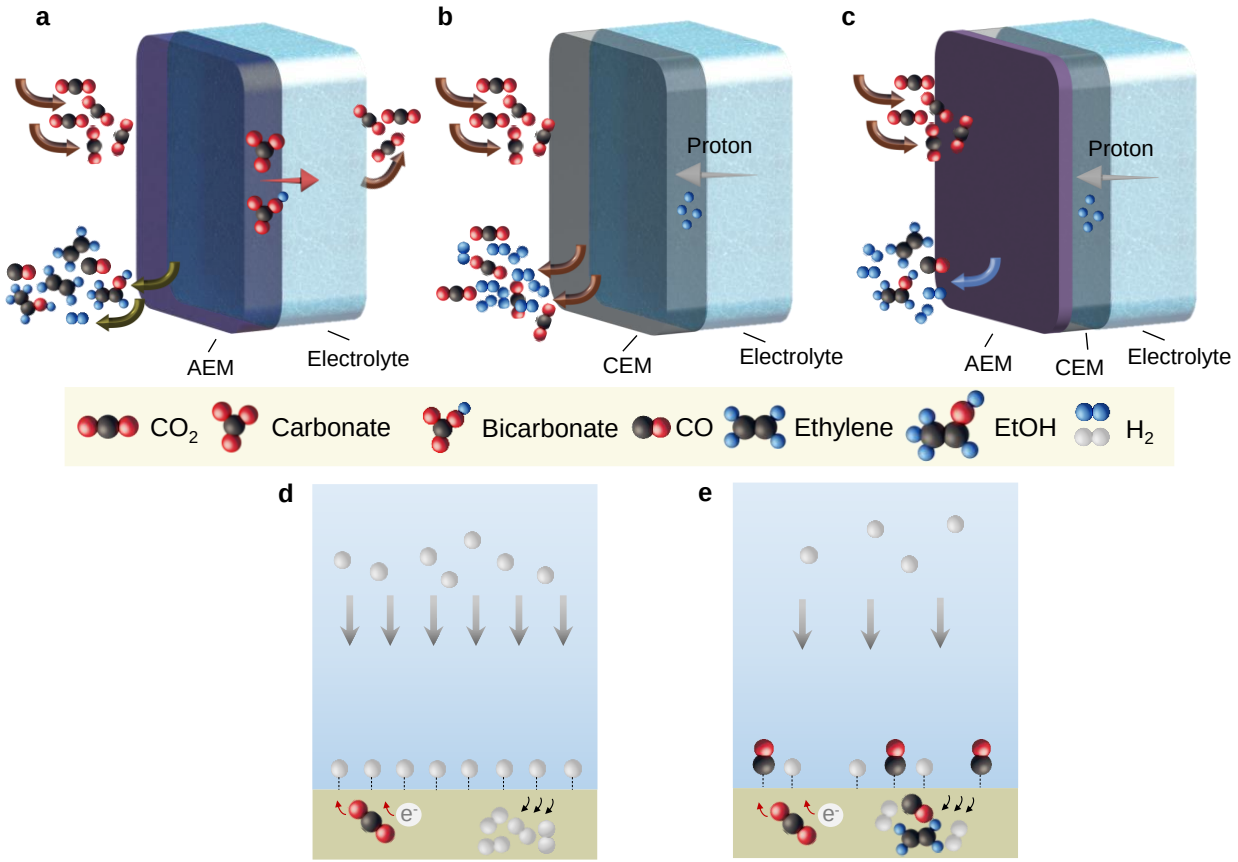
The (111) surface of face-centred cubic Cu was considered here using a $6 \times 3 \times 4$ periodic cell, since the (111) surface is found to be the dominant surface in the Cu-based electrocatalyst material utilized in this work, which is proved by the XRD measurements. 15 Å of vacuum space was set among the periodic surfaces in the z-direction to decouple the interaction between them. Ag-Cu surface was constructed by placing Ag clusters, including a 3-atom and 9-atom cluster, on the Cu(111) surface. Ag-doped Cu surface was constructed by replacing Cu atoms with Ag atoms. To reduce the computational cost, all atoms in the two bottom-most layers were fixed during the structural optimization calculations while other metal atoms and the adsorbates were allowed to relax.

The computational hydrogen electrode model[19] was used to calculate the Gibbs free energy by correcting the electronic energies directly determined from DFT calculations with Zero-point

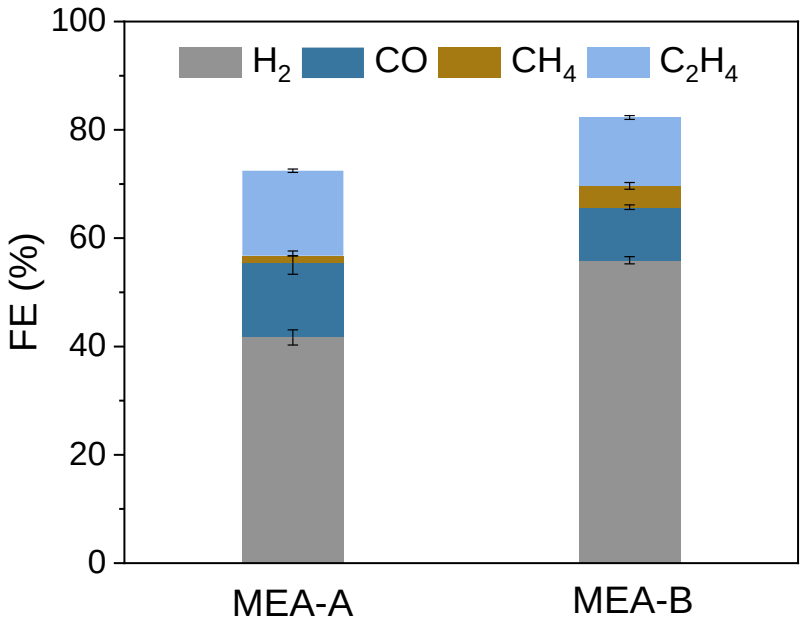
energies, entropies, and heat capacities, which are calculated from harmonic oscillator approximation at 298.15 K. Contributions to Gibbs free energies for each gas species and adsorbates are summarized in **Supplementary Table 7**. We note that the Gibbs free energies determined from our calculations provide a reasonable prediction of semiquantitative thermodynamic trends electrochemical conditions, since we ignore the presence of transition states and charged intermediates on surfaces.

Charge-density difference[20,21] was applied to investigate the charge transfer between the HCCOH* intermediate and surface. To consider the field and solvation effects, a hexagonal charged water overlayer-that is, five water molecules and one hydronium (H_3O^+)[22] – was introduced. To optimize the structure of the charged water overlayer, Ab initio molecular dynamics simulations were conducted in canonical ensemble (NVT) with the Nose-Hoover thermostat[23–25] and a 1.0 fs time step at 300 K, as performed in our previous study.[26] Reaction intermediates were included in the optimized geometry from AIMD simulations, and again to perform DFT calculations.

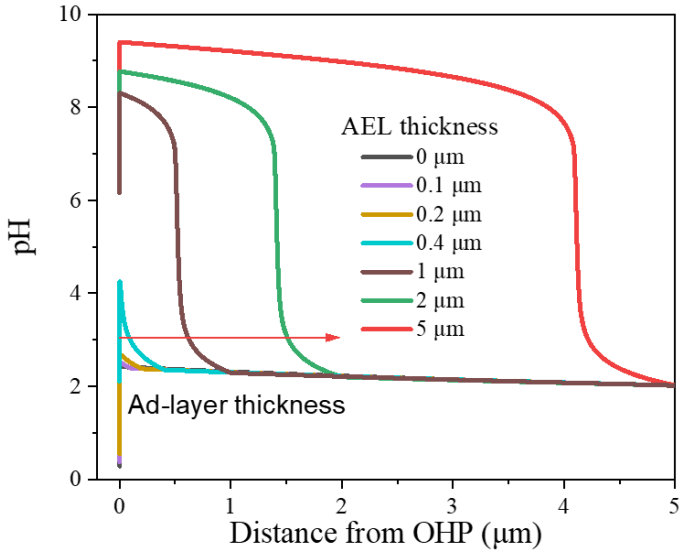
Supplementary Note 1 | Discussion on different MEA electrolyzer design and local microenvironment. MEA CO_2 electrolyzers with a cation exchange membrane (CEM) combined with acidic electrolyte were introduced to mitigate the CO_2 cross over issue (**Supplementary Figure 1**). In this configuration, the (bi)carbonate ion crossover would decrease, if not eliminated, due to the CEM-enabled acidity.[27] However, the unshielded cathode surface substantially favors the hydrogen evolution reaction (HER) over CO_2R due to the high availability of protons and the fast kinetics of the HER (**Supplementary Figure 1d**).[27,28] Using a bipolar membrane in a forward-bias bipolar membrane (f-BPM) mode (**Supplementary Figure 1c**) in a MEA enables high carbon efficiency and CO_2R.[27,29] In this configuration, the AEM would maintain a local alkaline environment and enable (bi)carbonate ion transport towards the CEM, while the CEM provides protons. Due to carbonate equilibria, the (bi)carbonate ions react with protons to regenerate CO_2 and H_2O between the AEM and CEM, which tremendously decreases the CO_2 loss to the anode.[27,28] By depositing an AEM ad-layer on the catalyst surface, the local pH would increase and favor CO_2R (**Supplementary Figure 1e**).[27]



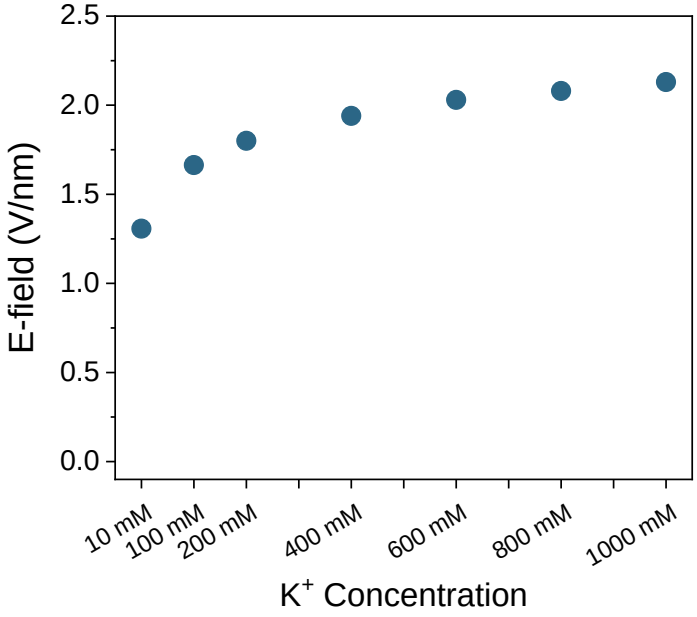
Supplementary Figure 1. Different configurations of CO_2 MEA electrolyzers. (a) MEA with an AEM membrane which results in >70% CO_2 crossover to anode via (bi)carbonate. (b) MEA with a CEM membrane that minimizes CO_2 crossover (< 30%). (c) MEA with a forward bias-BPM that reduces the CO_2 crossover with a basic local cathode surface. (d) Microenvironment in an MEA with a CEM membrane. The catalyst surface is expected to be too acidic due to proton accumulation at the outer Helmholtz plane (OHP). Due to the fast kinetic nature of HER, these accumulated protons can be easily reduced to H_2 by the electron on the electrode. (e) Microenvironment in a f-BPM MEA which favors CO_2RR by providing local alkalinity.



Supplementary Figure 2. CO_2R performance of Cu catalyst coated with benzimidazolium (an anion exchange ad-layer) in MEA. CO_2R performance at different anolyte concentration in MEA: (MEA-A) 0.01 M H_2SO_4 and (MEA-B) 0.2 M H_2SO_4. CO_2R was performed at 200 mA cm^{-2} for these experiments. The anion exchange layer coating is 3 mg cm^2. The plotted values are means of three replicates, while error bars indicate standard deviation.

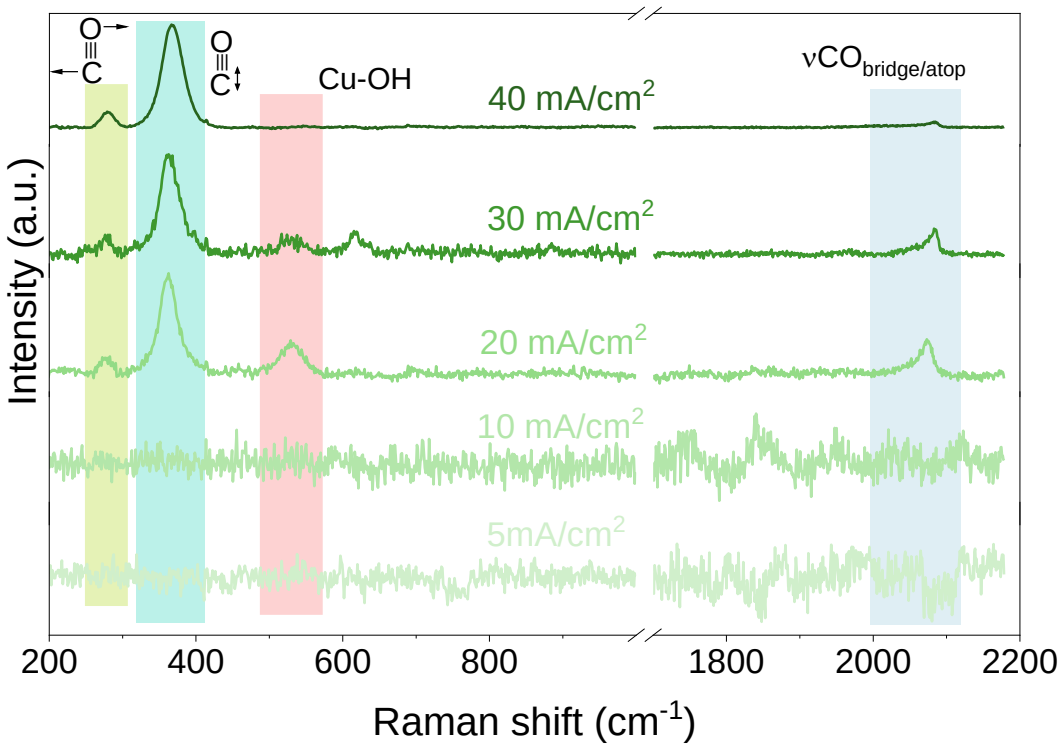


Supplementary Figure 3. Modelling results of local pH in the presence of anion exchange layer. Modeled pH changes away from the outer Helmholtz plane (OHP) with different anion exchange layer thicknesses under an applied current density of 100 mA cm^{-2}. If no anion exchange layer was used, the catalyst surface is expected to be too acidic due to proton accumulation at the outer OHP (as schematically shown in Supplementary Figure 1d). Due to the fast kinetic nature of HER, these accumulated protons can be easily reduced to H_2 by the electron on the electrode. However, by depositing an anion exchange layer on the catalyst surface, the local pH would increase. More alkalinity is expected as a thicker anion exchange layer is deposited which favors CO_2R (Supplementary Figure 1e).

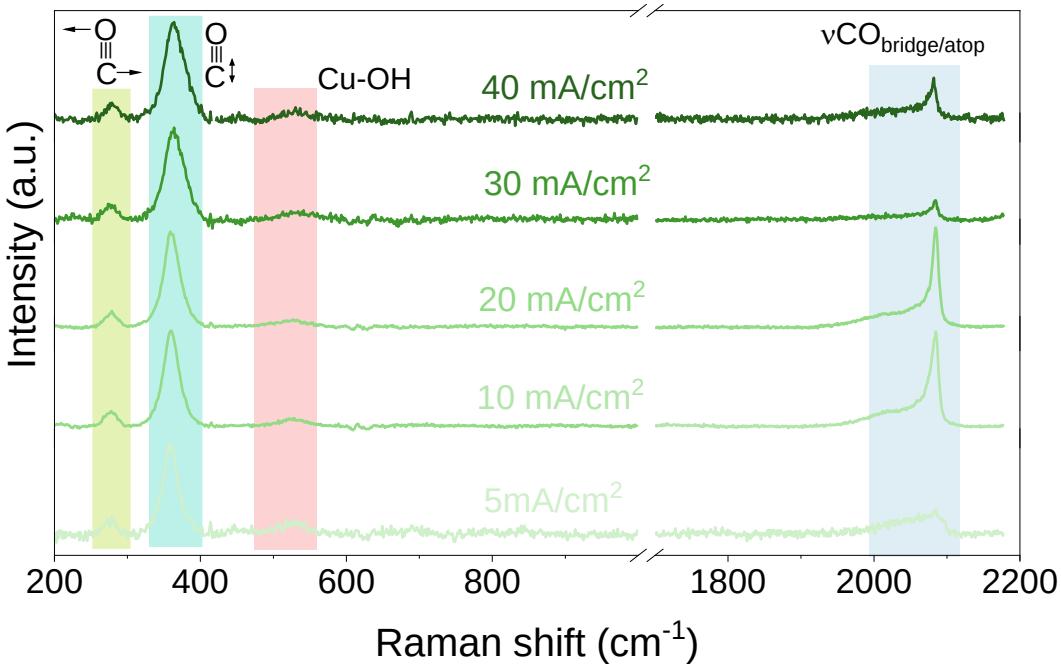


Supplementary Figure 4. Effect of cation concentration on the electric field at the cathode surface. This modeling result shows a small amount of cation can increase the cathode surface electric field and facilitate CO_2R.

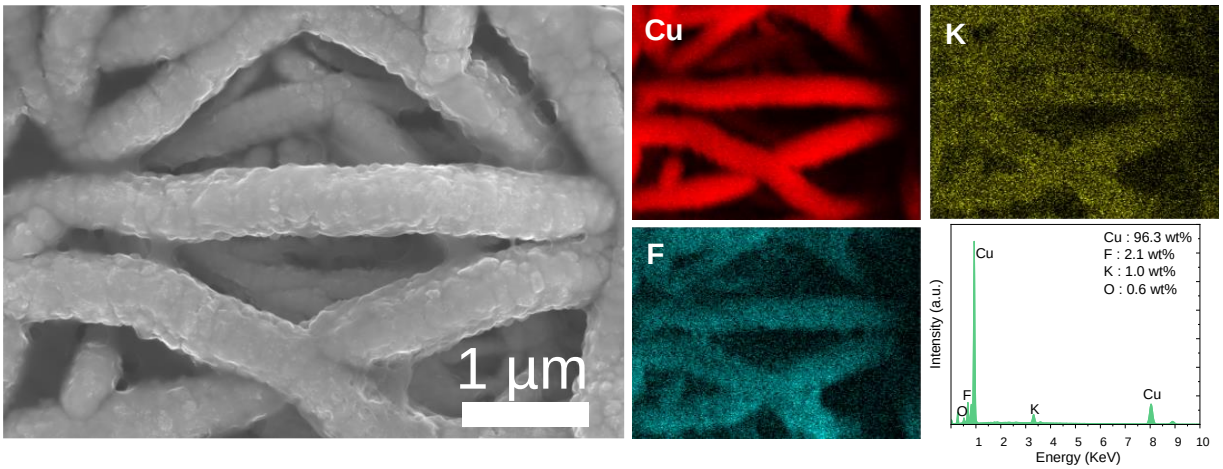
Supplementary Note 2 | Discussion on Surface-Enhanced Raman Spectroscopy – cation effect. To investigate the influence of alkali cation on acidic CO_2R efficiency, we incorporated 5 mM K_2SO_4 to the electrolyte of Raman cell. The CO intermediate is typically detected using surface-enhanced Raman scattering (SERS) in three distinct Raman shift regions: 280-290 cm^{-1} for the restricted rotation of adsorbed CO, 360-370 cm^{-1} for Cu-CO stretching, and 1800-2100 cm^{-1} for C-O stretching. The Raman shift within the 520-530 cm^{-1} range can be attributed to the Cu-OH band. It has been shown that a stronger intensity of the Cu-CO stretching band corresponds to a higher coverage of the CO intermediate and increased C_{2+} selectivity. Incorporating an alkali cation into the electrolyte results in a noticeable enhancement in the intensity of the restricted rotation of adsorbed CO and Cu-CO stretching, indicating a greater CO coverage. This promotes C-C coupling and C_{2+} formation (**Supplementary Figure 5 and 6**). The results indicate that the introduction of small amount of alkali cation close to the catalysts surface is critical to facilitate CO_2R. This result is an agreement with previous experimental observations in other configurations.[29–31] For example, Yang et al. observed an increase in CO FE as the salt concentration increased in a reverse BPM MEA with a silver-based cathode.[29] However, this trend may coincide with salt formation. Although this approach succors CO_2RR and suppresses HER compared to a salt-free electrolyte, the salt formation on the cathode limits the stable performance of the electrolyzer. Thus, we pursued strategies to enrich the catalyst surface with alkali cations while using a salt-free acidic anolyte.



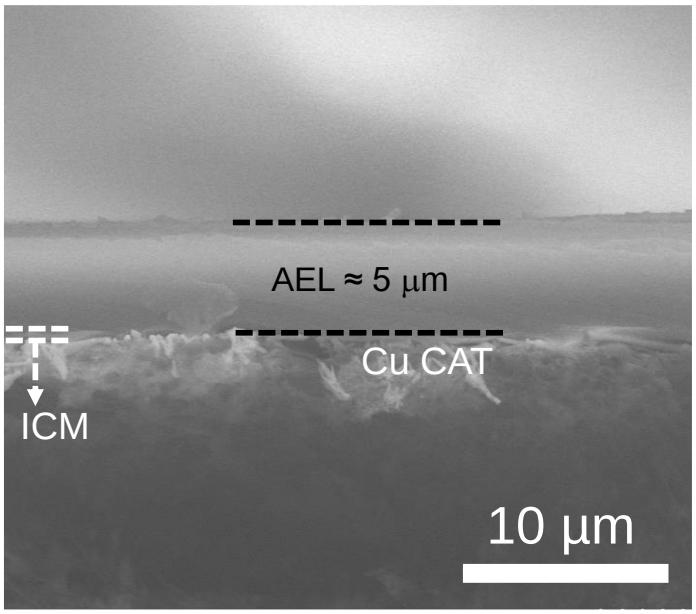
Supplementary Figure 5. *Operando* Raman analysis of Cu catalyst in acidic electrolyte. Assignment of the Operando Raman spectroscopy features of Cu electrode at different current densities in contact with 0.01 M H_2SO_4: adsorbed CO (*CO, 280-290 cm^{-1}) frustrated rotation, stretching of Cu-CO (360-370 cm^{-1}), Cu hydroxides (Cu-OH, 520-530 cm^{-1}), bridge-bound CO and atop-bound CO on Cu (Cu-CO_{bridge} and Cu-CO_{atop}, 1800-2100 cm^{-1}).



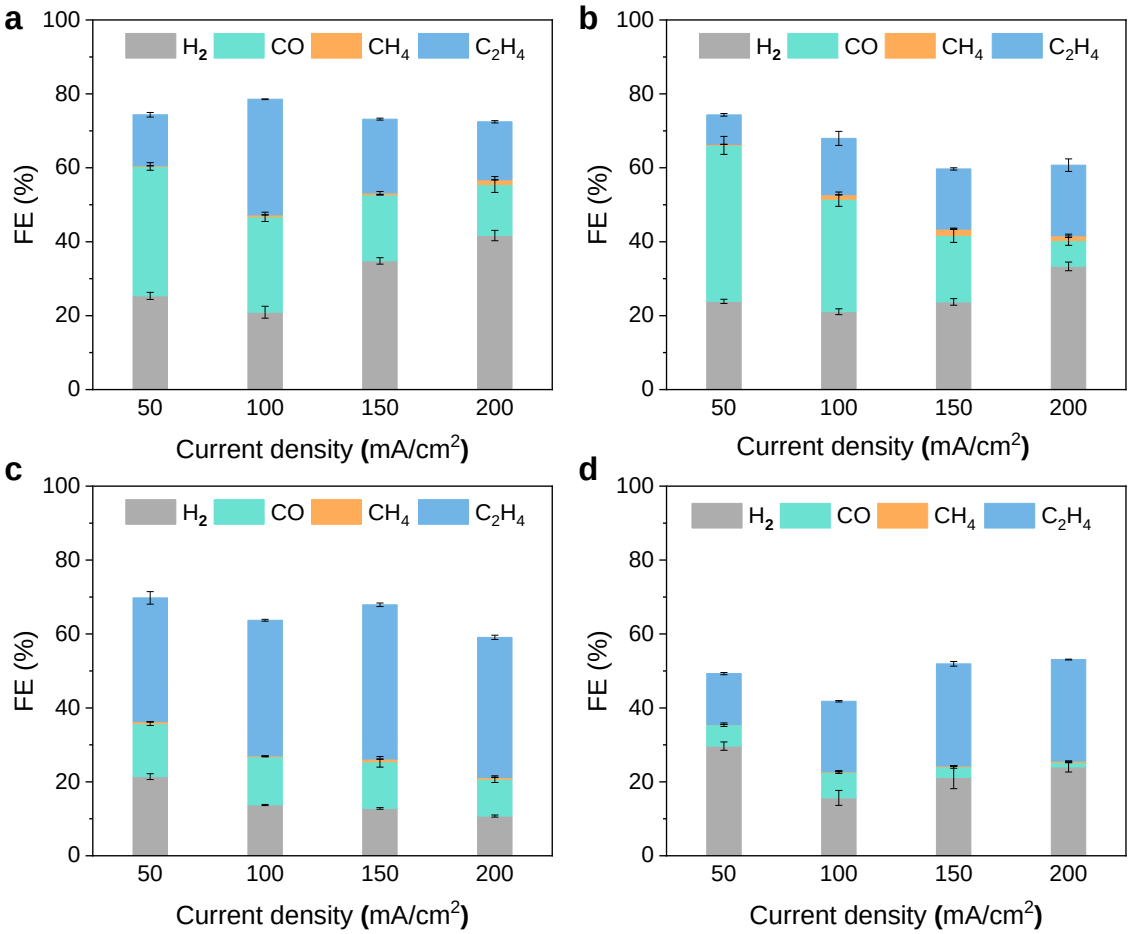
Supplementary Figure 6. *Operando* Raman analysis – effect of cation. Assignment of the Raman features (operando spectrum of Cu at different current densities) in contact with 0.01 M H_2SO_4 + 5 mM K_2SO_4. The results show that adding cation to the electrolyte increases the CO coverage that can facilitate CO_2R.



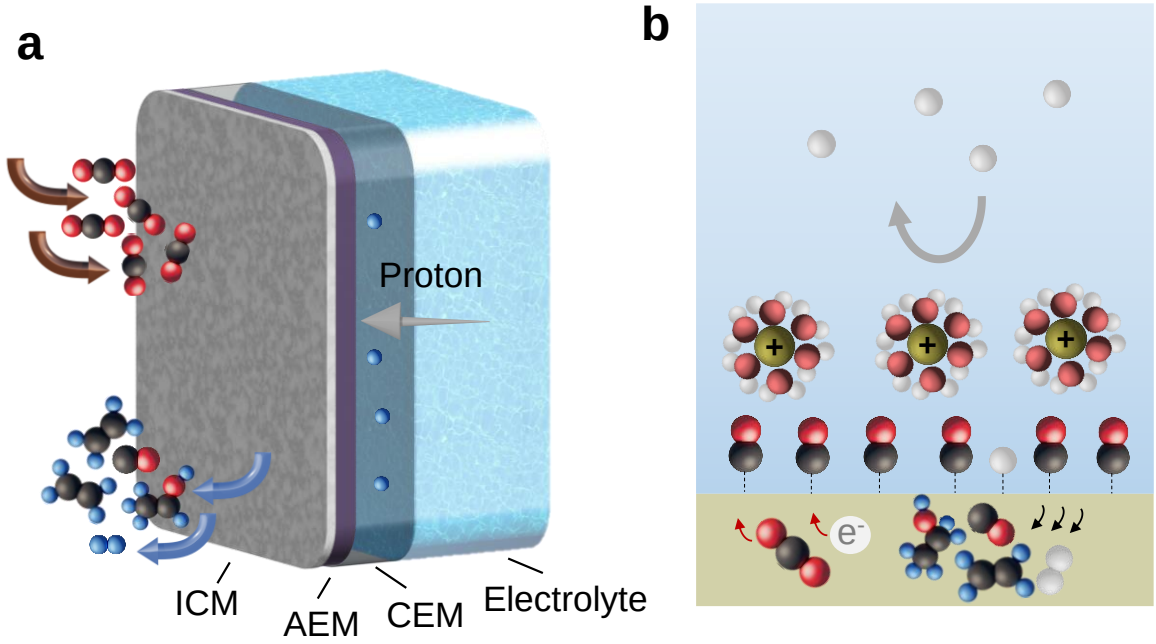
Supplementary Figure 7. SEM characterization of Cu catalyst coated with ICM. SEM images of Cu/PTFE coated with ICM with the corresponding Cu, F, and K EDS elemental mapping showing a homogeneous distribution of potassium over the catalyst surface.



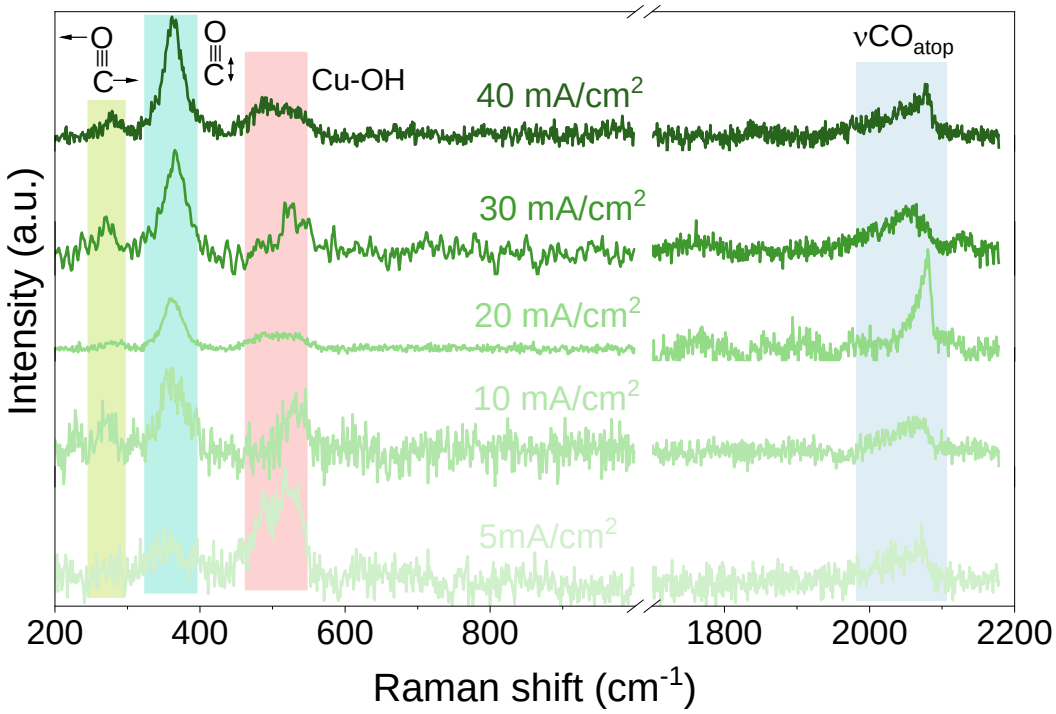
Supplementary Figure 8. SEM of anion exchange layer. Cross-sectional scanning electron microscopy image of gas diffusion electrode equipped with ICM and AEL. The AEL thickness is about 5 μm.



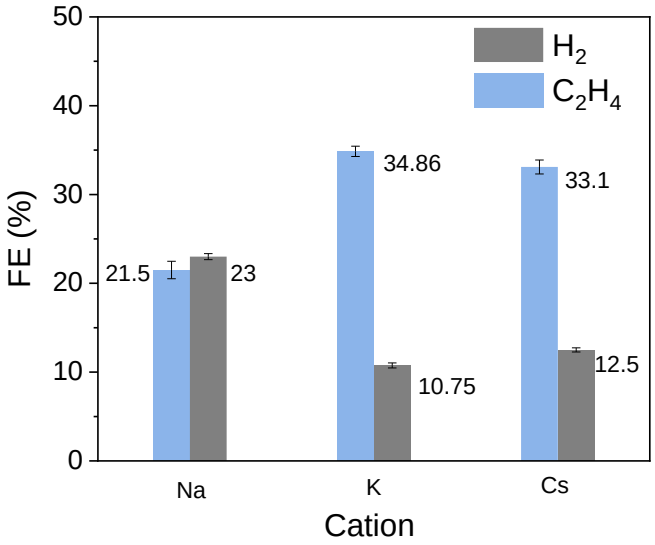
Supplementary Figure 9. Effect of ICM loading on CO_2R performance. Gas product FE for a Cu catalyst with different loading cation in the ICM layer: (a) 0, (b) 2, (c) 4, and (d) 8 µl/cm^2. The AEL loading was similar (3 mg/cm^2) for these experiments. The ratio of cation (K^+) to PFSA loading was 2. The case where there is zero cation loading indicates the absence of ICM. The plotted values are means of three replicates, while error bars indicate standard deviation.



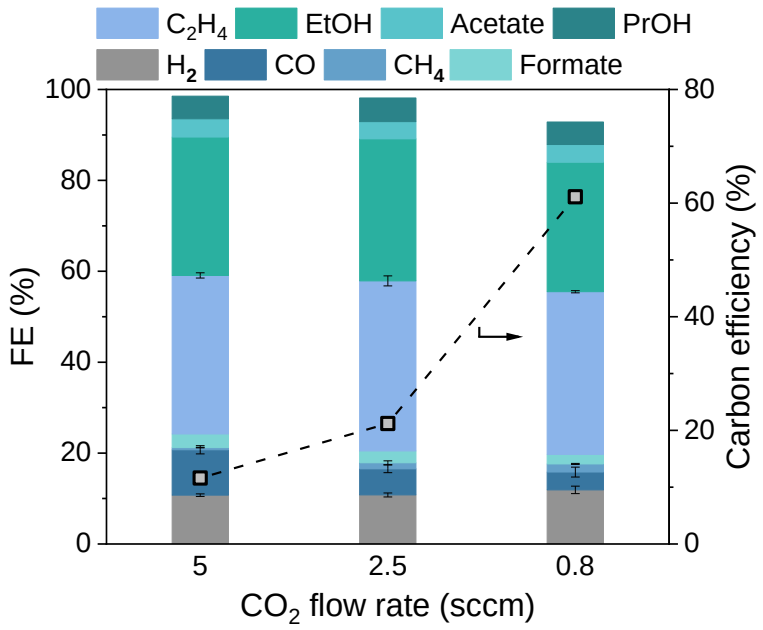
Supplementary Figure 10. Schematic of electrolyzer configuration and EDL for f-BPM MEA electrolyzer equipped with ICM. (a and b) ICM and AEL (AEM) ad-layers on the MEA electrolyzer. The ICM strategy can provide cation on the cathode surface and increase the local pH, improving CO_2 electroreduction.



Supplementary Figure 11. *Operando* Raman analysis of Cu catalyst equipped with ICM. Operando Raman spectroscopy studies of Cu catalyst equipped with ICM that indicates a higher intensity for the CO coverage peaks. Higher CO coverage facilitates C—C coupling and C_{2+} production.



Supplementary Figure 12. Effect of alkali cation type in the ICM on the CO_2R performance. Hydrogen and ethylene FE of ICM-modified PTFE–Cu electrodes with different cations inside the ICM layer at a concentration of 4 μl/cm^2. All experiments were performed at 200 mA/cm^2. The plotted values are means of three replicates, while error bars indicate standard deviation.

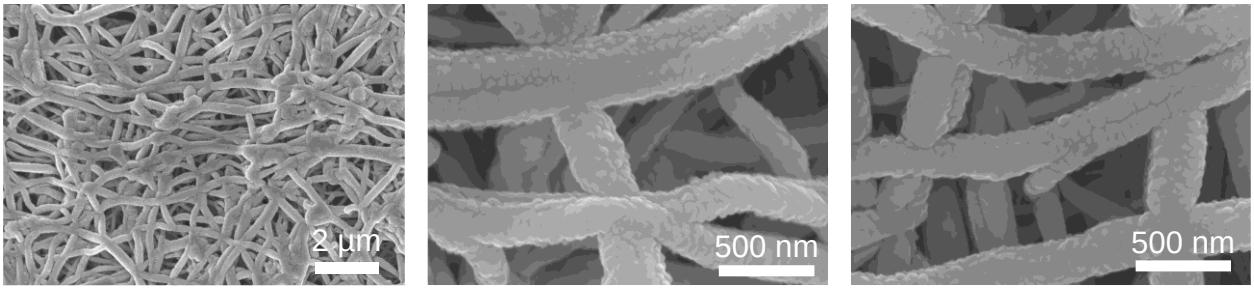


Supplementary Figure 13. Carbon efficiency of Cu catalyst equipped with ICM. CO_2R product distribution and carbon efficiency (dash line) of CO_2 to all products on an ICM-modified PTFE–Cu electrode at various CO_2 flow rates at 200 mA/cm^2. The plotted values for gas products are means of three replicates, while error bars indicate standard deviation.

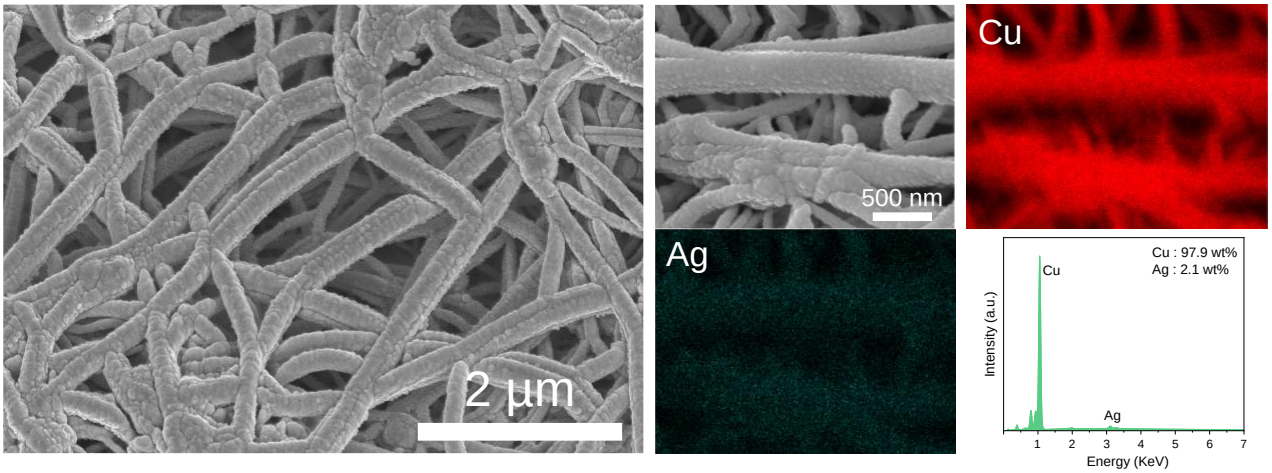
Supplementary Note 3 | Discussion on energy efficiency/full-cell voltage reduction. To allow CO_2 electrolyzers to be closer to industrial valorization, the electrolyzer energy consumption must be decreased. The current largest contributor to the total electrolyzer energy cost is the energy spent as the electricity – or cell potential – necessary to run the electrolyzer at industrially relevant current densities.[32,33] Thus, we focused our attention on improving the electrolyzer energy efficiency at an industrially relevant current density (200 mA/cm^2). We employed electrochemical Impedance Spectroscopy (EIS) to measure the IR drop across the MEA electrolyzer and identify the main sources of ohmic losses. Previous acid-fed CO_2 electrolyzers employed Nafion 117 as the CEM.[27,2834,35] We began our analysis performing CO_2R in MEA electrolyzer equipped with Nafion 117. By performing this analysis, through the Nyquist plots, we observed that ohmic losses are significantly high and increase a lot as we increase the current density (**Supplementary Table 8**). Two main sources of ohmic losses are membrane and contact resistance of the membrane/catalyst.

We then targeted ohmic overpotential reduction by using thinner CEMs. We tested three different types of CEM membrane: Nafion117, Nafion 212, and Nafion XL. Using a thinner membrane reduced the full-cell voltage with a larger reduction at higher current density; the applied cell potential decreased from 4.6 V to 4.0 V at 200 mA/cm^2 when Nafion 117 (183 µm) and Nafion XL (27 µm) were used, respectively (**Supplementary Figure 28**). We sought to further decrease the applied cell potential by pursuing the ohmic resistance caused by the contact between the anode and the membrane (**Supplementary Figure 28**). To decrease this resistance, we prepared a catalyst coated membrane (CCM) by spray-coating anode material (IrO_2 nanoparticles) on the CEM.[36] By using a thinner membrane such as Nafion XL-CCM instead of Nafion 117, a savings of 1.1 V in the full-cell voltage at 200 mA/cm^2 can be achieved which is of great

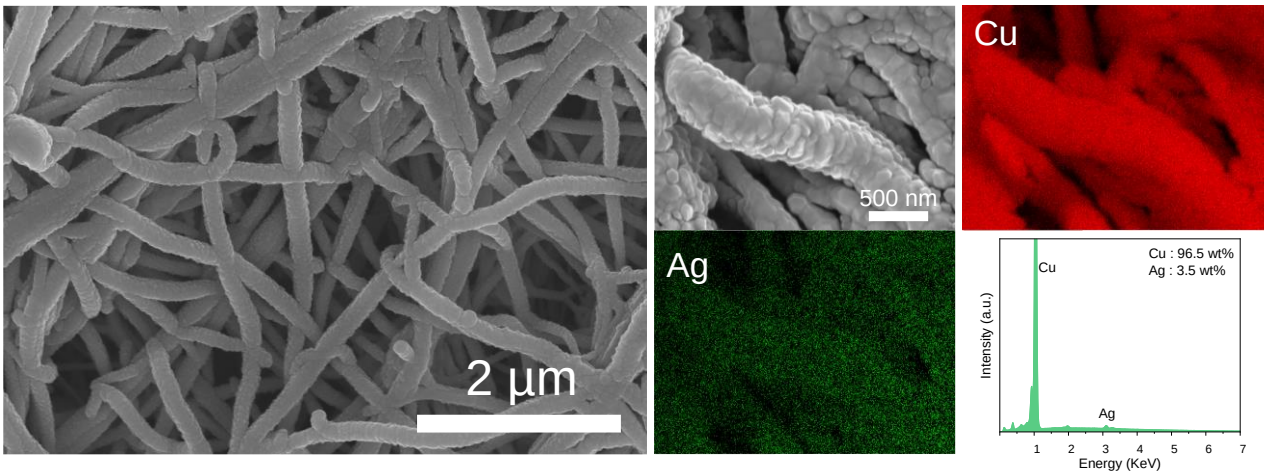
importance for commercial CO_2R viability. The CCM approach was utilized for the remainder of this work.



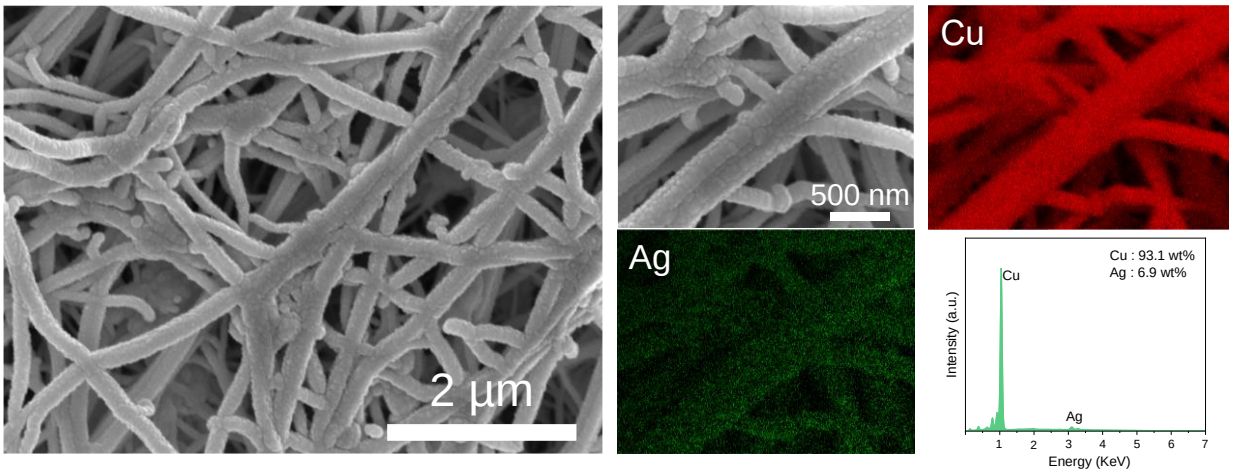
Supplementary Figure 14. SEM characterization of Cu catalyst. SEM images of Cu/PTFE at various magnifications.



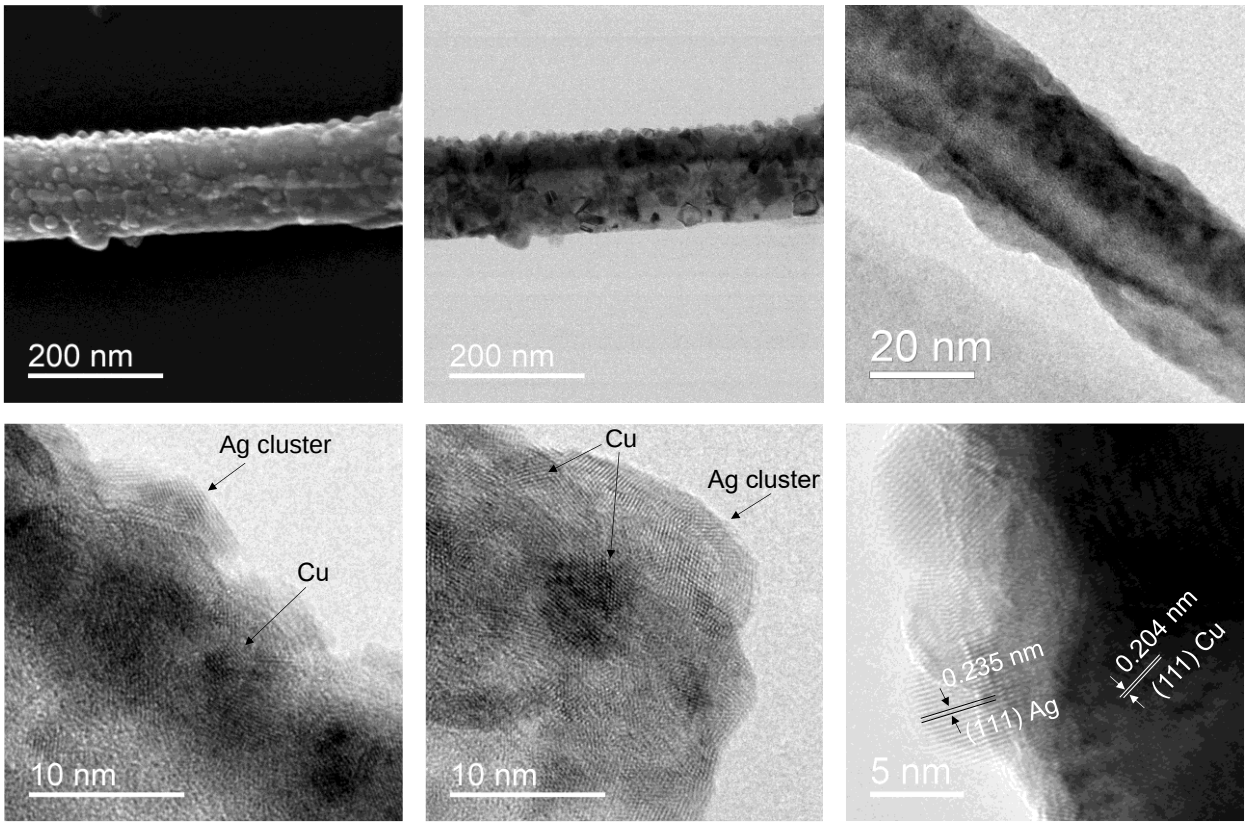
Supplementary Figure 15. SEM characterization of Cu-2 nm Ag catalyst. SEM images of Cu-2 nm Ag/PTFE at various magnifications with the corresponding Cu, and Ag elemental mapping, demonstrate a homogeneous distribution of Ag across the Cu surface. The nominal loading of Ag was 2 nm via sputtering.



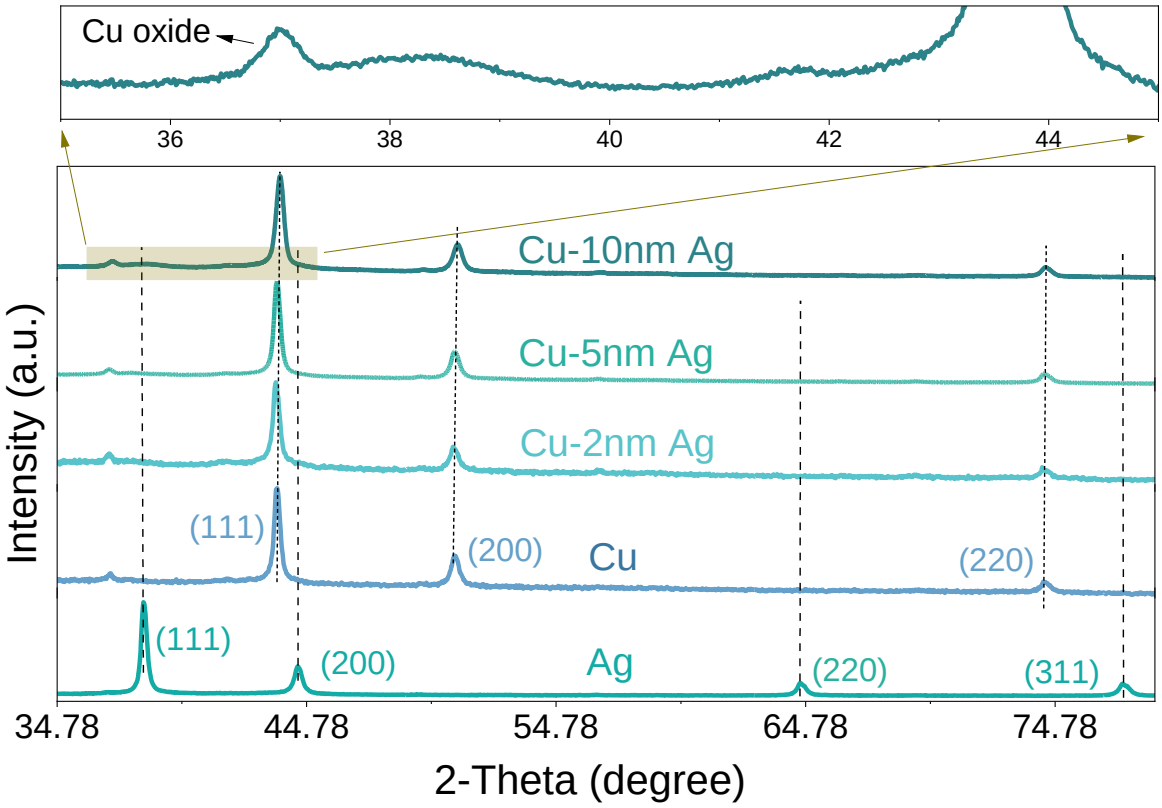
Supplementary Figure 16. SEM characterization of Cu-5 nm Ag catalyst. SEM images of Cu-5 nm Ag/PTFE at various magnifications with the corresponding Cu, and Ag elemental mapping, demonstrate a homogeneous distribution of Ag across the Cu surface. The nominal loading of Ag was 5 nm via sputtering.



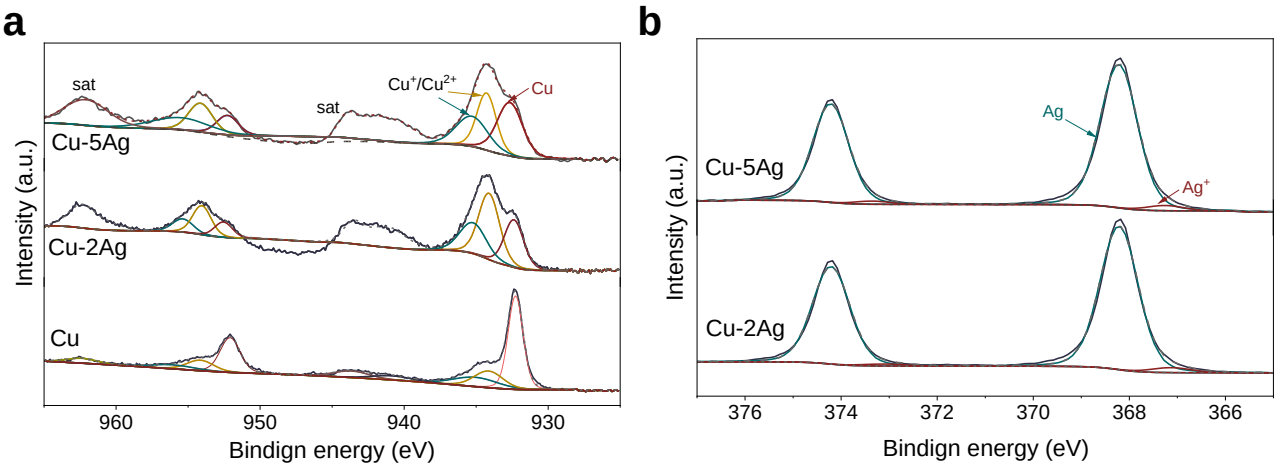
Supplementary Figure 17. SEM characterization of Cu-10 nm Ag catalyst. SEM images of Cu-10 nm Ag/PTFE at various magnifications with the corresponding Cu, and Ag elemental mapping, demonstrate a homogeneous distribution of Ag across the Cu surface. The nominal loading of Ag was 10 nm via sputtering.



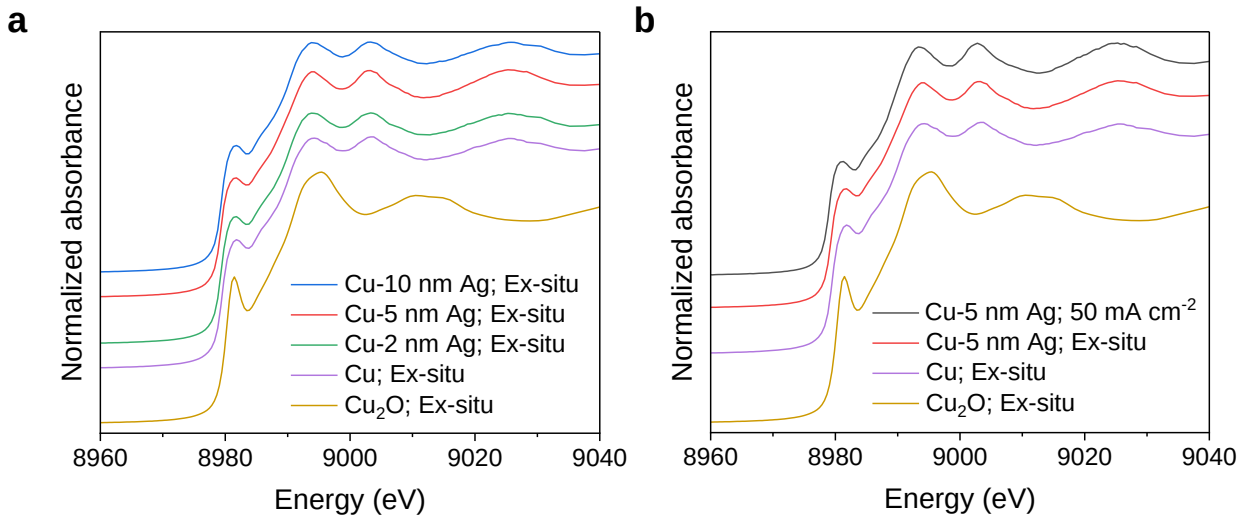
Supplementary Figure 18. STEM and HRTEM images of Cu- 5 nm Ag catalyst at different magnifications. HRTEM images indicate that the deposition of the Ag layer results in a nanocluster structure rather than a compact layer of Ag on the Cu surface. This could be attributed to the porous nature of the Cu-PFTE (SEM images on Supplementary Figure S13-S16) and the limited thickness of the Ag layer (5 nm), which prevented complete coverage of the Cu surface.



Supplementary Figure 19. XRD patterns of catalysts. X-ray diffraction patterns of PTFE-Ag, PTFE-Cu, and PTFE-Cu-Ag electrodes with different loadings of silver (2 nm, 5 nm, and 10 nm). No clear silver peak was observed in Cu-2 nm Ag and Cu-5 nm Ag catalysts, which can be attributed to the small Ag cluster size; however, XRD reveals a subtle, less crystalline Ag peak in the Cu-10 nm Ag catalyst.

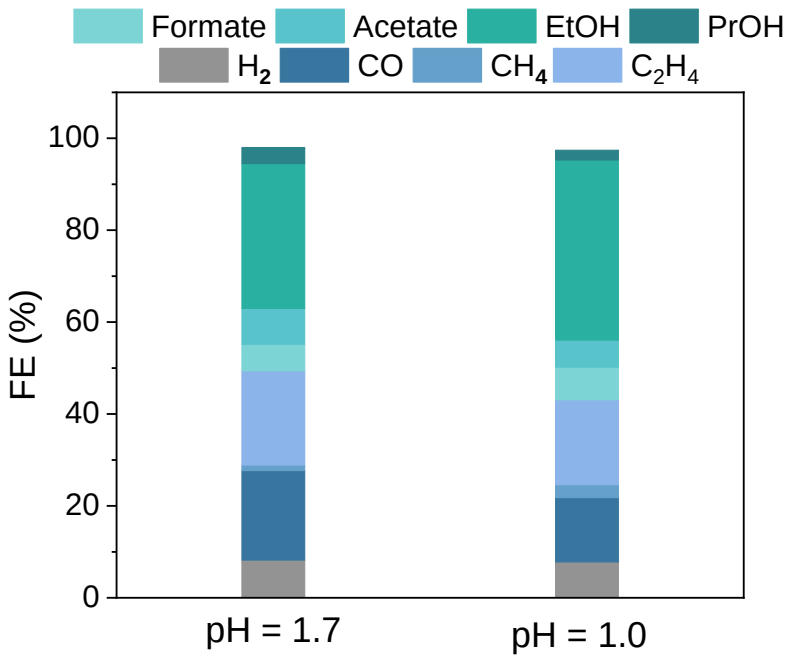


Supplementary Figure 20. High-resolution XPS spectra. (a) Cu2p and (b) Ag3d. The XPS analyses clearly shows the presence of Ag on the surface of Cu.



Supplementary Figure 21. Ex-situ and in-situ X-ray absorption spectroscopy (XAS) spectra. (a) Ex-situ XAS analyses for the Cu-Ag catalysts reveal that bulk Cu primarily exists in its metallic form (Cu^0). The ex-situ spectra were recorded in transmission mode. (b) In-situ and ex-situ XAS analyses for the Cu-5 nm Ag catalyst reveal that during CO_2R, the Cu primarily exists in its metallic form (Cu^0). The in-situ XAS spectra of the Cu K-edge were recorded in fluorescence mode using a custom in-situ XAS flow cell at a constant current of 50 mA cm^{-2}.

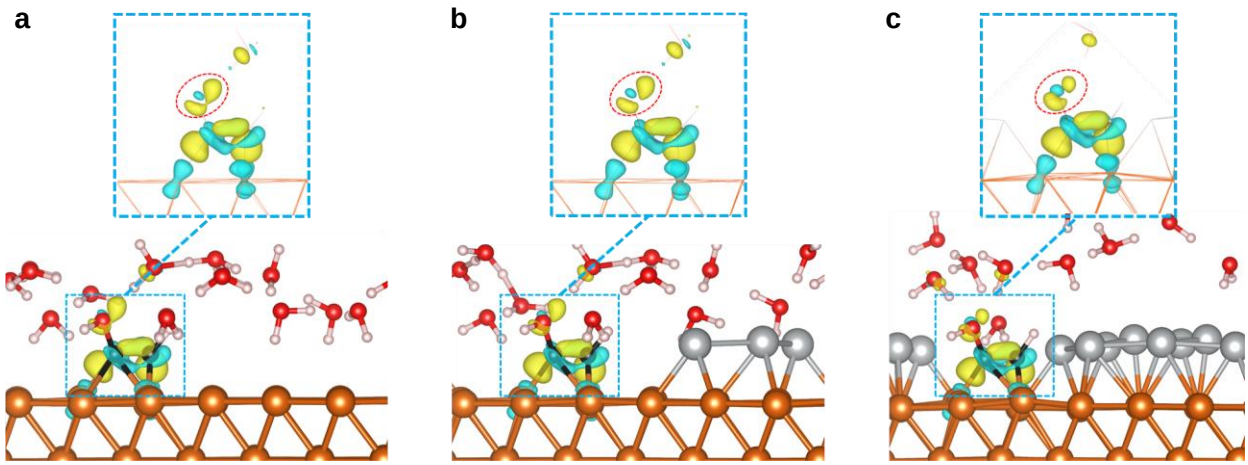
The ex-situ spectra show the copper to be predominantly in the metallic $Cu^{(0)}$ state. All samples retain their $Cu^{(0)}$ state under reaction conditions, as evidence by the unchanged spectra during in-situ characterization.



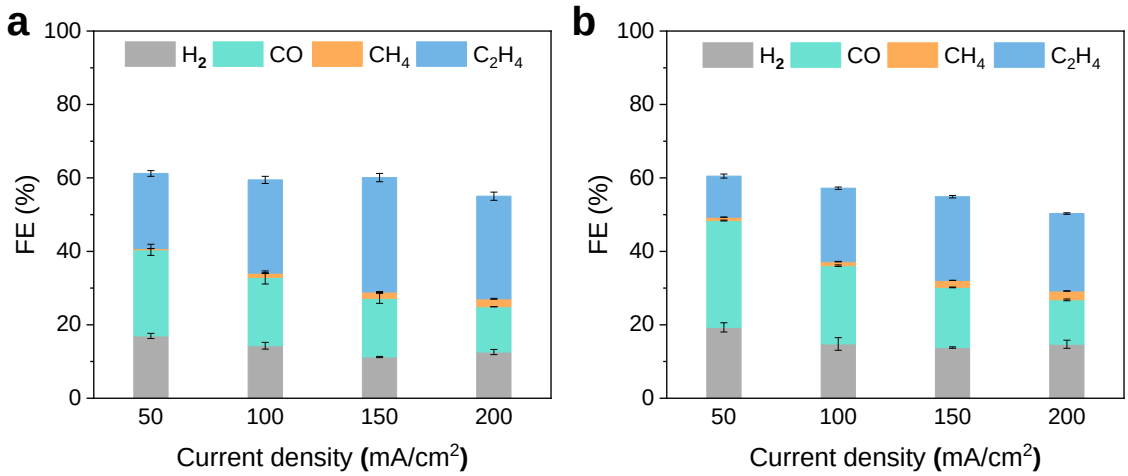
Supplementary Figure 22. Effect of pH on ethanol production. CO_2R product distribution for a Cu-5Ag catalysts at 300 mA/cm^2 with different electrolyte. The experiment was performed in a flow cell to study the effect of pH on ethanol and ethylene formation. The electrolytes were 0.01M H_2SO_4 + 3M KCl (pH = 1.7) and 0.05M H_2SO_4 + 3M KCl (pH = 1.0). This experiment shows that increasing the H^+ concentration (lower pH) facilitates ethanol formation in acidic CO_2R.

Supplementary Note 4 | DFT modelling of Cu-Ag interface for CO_2R. Previous studies have established Ag as a catalyst for producing CO from CO_2, while Cu facilitates C-C coupling, leading to formation of C_{2+} products. In our Cu-Ag system, Cu promotes the production of C_{2+} products by coupling the CO generated by Ag clusters on the Cu surface. Therefore, the Cu-Ag interface, rather than the Ag cluster itself, plays a critical role in determining the reaction selectivity toward C_{2+} products. To gain mechanistic insight into the effect of Ag on the formation of C_2 intermediates, we conducted DFT calculations to investigate the Cu-Ag interface on Cu surfaces. First, AIMD simulations were conducted to optimize the hexagonal charged water layer on Cu-Ag surface, comprising five water molecules and one hydronium (H_3O^+), to account for the field and solvation effects. Although alkali cations were not included in this model, it still provides a reliable framework for investigating the reaction energy profile of C_2 intermediate formation with reasonable computational accuracy and speed, as suggested by Montoya et al.[37]

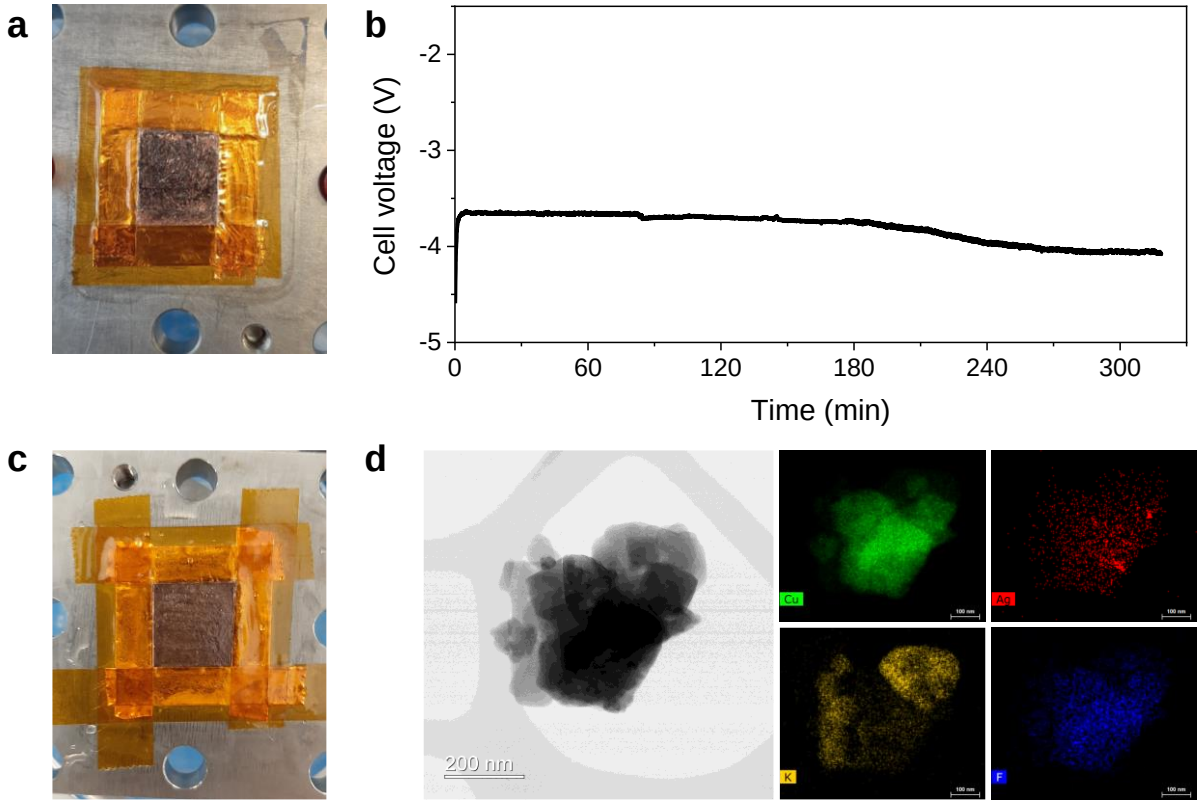
Next, DFT calculations were applied to study the key reaction intermediates involved in ethanol and ethylene formation, such as $HCOOH^*$, $HCCHO^*$, H_2CCO^* on the Cu-Ag surface optimized by AIMD. Our DFT calculations revealed that the Cu-Ag interface indeed favours the formation of C_2 intermediates on the path to ethanol production compared to Cu surface, providing valuable mechanistic insights into the Cu-Ag catalyst's superior ethanol production performance.



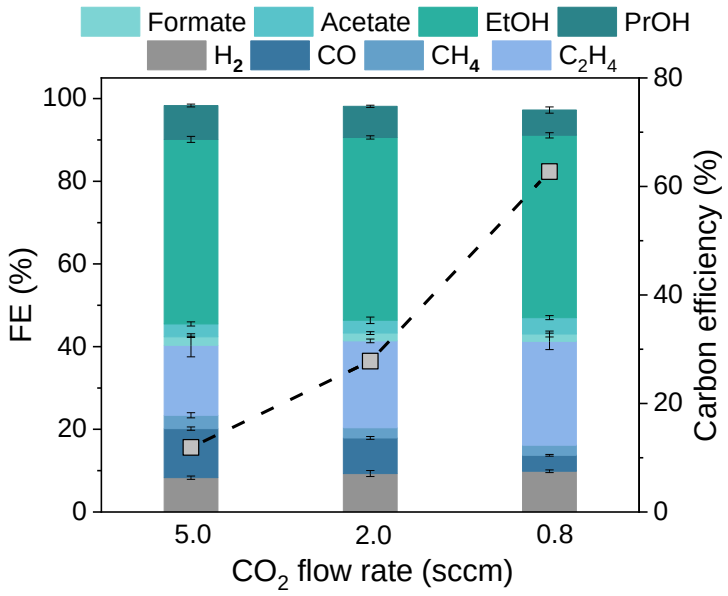
Supplementary Figure 23. Electron density difference plots for Cu and Cu-Ag catalysts. Surface configurations of (a) Bare Cu, (b) Cu with 3 atoms of Ag on top, and (c) Cu with 9 atoms of Ag on top (which create an Ag cluster through which C–O bond scission is thermodynamically unfavourable on the interface of Cu-Ag clusters). Ag clusters on Cu sites would increase the local concentration of CO. Thereby, a local CO reservoir is created. The availability of CO would promote ethanol production on the Cu sites by stabilizing the C–O bond and H* spillover mechanism.



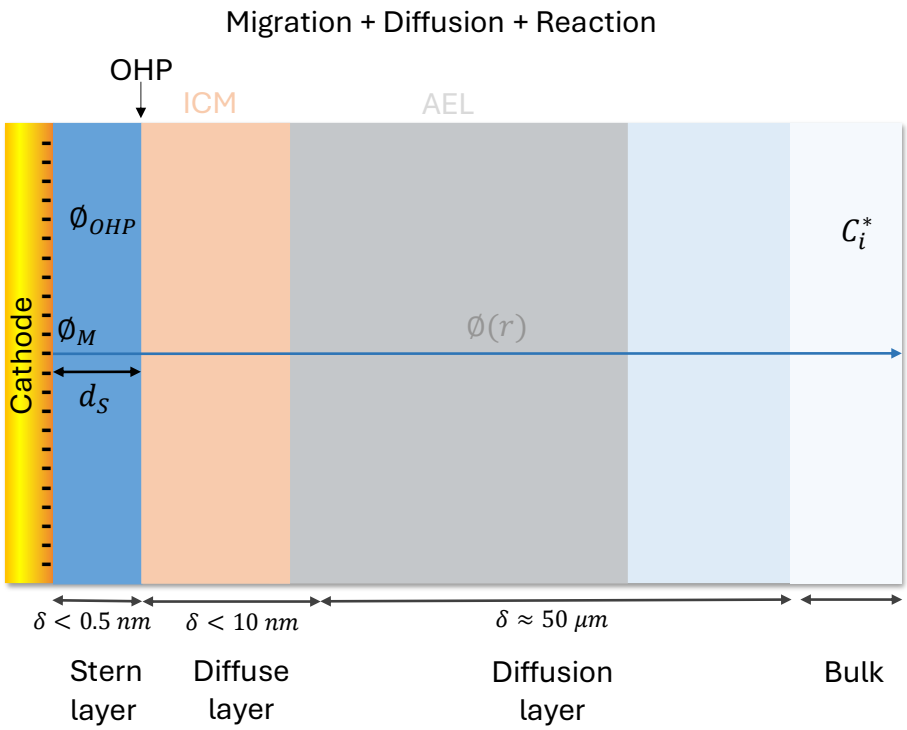
Supplementary Figure 24. Effect of deposition of AgNP on CO_2R performance. CO_2R product distribution for a PTFE/Cu catalyst with Ag nanoparticles (AgNPs, 20 nm) coating, corresponding to sputtering loadings of 5 nm and 10 nm. The experiments demonstrate that adding AgNPs to the Cu catalyst increases CO FE, which correlates with AgNP acting as a CO catalyst. In this work, sputtering was employed for Ag deposition to access very small Ag nanoclusters (1-10 nm loading), thereby increasing the Cu-Ag interfaces and ethanol production. The plotted values are means of three replicates, while error bars indicate standard deviation.



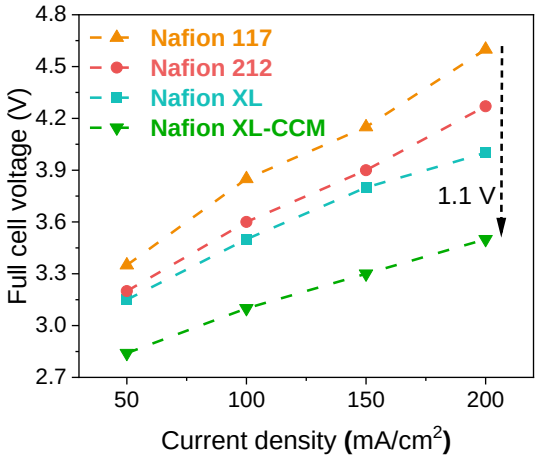
Supplementary Figure 25. Stability of Cu-5Ag catalyst with and without the ICM adlayer. (a) Image of post-electrolysis catalyst surface without the ICM adlayer. (b) Stability of the Cu-Ag without the ICM adlayer. It shows that the ICM layer enhances the stability of the catalyst (Figure 4). (c) Image of post-electrolysis catalyst surface with the ICM adlayer after the stability test. (d) STEM image and corresponding EDS mapping of a Cu-Ag catalyst equipped with the ICM adlayer after stability testing. The images and mapping confirm the presence of ICM (K and F) after CO_2 electrolysis.



Supplementary Figure 26. Carbon efficiency of a Cu-Ag catalyst. The carbon efficiency of a Cu-Ag (with a loading of 5 nm) catalyst equipped with ICM as a function of CO_2 flow rate at 200 mA/cm^2. The plotted values are means of three replicates, while error bars indicate standard deviation.



Supplementary Figure 27. **Multiphysics simulation domain.** 1D domain for the simulation of space charge density, E-field, and pH.



Supplementary Figure 28. Full-cell voltage for acid-fed CO_2 MEA electrolyzer with different CEM membranes. Using a thinner membrane with a catalyst coating (CCM; Nafion XL-CCM *vs* Nafion 117) could save 1.1 V in the full-cell voltage at 200 mA cm^{-2}.

Supplementary Table 1. Total energy cost comparison for ethanol production. Comparison of total energy cost for different CO_2 to ethanol electrolyzers. The comparison is limited to electrolyzers working at total current densities ≥ 100 mA/cm^2

Parameters	**Alkaline CO_2R**			**Neutral CO_2R**				**Acidic CO_2R**						
Cell voltage (V)	2.8	3	3	3.9	3.7	3.5	3.7	3.8	3.75	3.5	4.2	3.8	3.7	3.5
EtOH FE (%)	26	40	69	23	41	50	52	20	21	12	12	9	16	44
Current Density (mA/cm^2)	100	900	271	300	600	190	125	200	100	500	1200	100	200	200
Carbon efficiency (%)	4	5*	5*	15*	15*	7	5	50	40	70	70	85	75	63
EtOH conc. (wt.%)	1	1	1	4	1	1	1	13	4	1	1	1	1	1
EtOH crossover (%)	60%	60%	60%	50%	60%	80%	80%	60%	20%	60%	60%	60%	60%	60%
Electrolyzer specific energy distribution (GJ (ton product)$^{-1}$)														
CO_2 Capture	45	29	23	62	37	23	23	74	60	105	105	93	92	28.2
Electrolyzer electricity	271	188	110	426	227	176	179	478	449	733	880	1061	589	199.9
Ethanol distillation	22	22	22	16	17	22	22	15	22	22	22	22	22	22
Cathode gas separation	578	285	235	118	75	146	220	45	55	28	28	10	19	10.2
Anode gas separation	88	57	33	100	56	46	44	-	-	-	-	-	-	-
Total energy cost (GJ/ton EtOH)	**1004**	**581**	**423**	**723**	**412**	**413**	**487**	**612**	**586**	**887**	**1034**	**1187**	**721**	**260.3**
Ref.	38	39	40	41	42	43	44	45	28	26	30	46	31	**This work**

*These are not experimental values; the maximum theoretical carbon efficiency was considered for the energy cost calculation.

Supplementary Table 2. Ethanol FE in the recent literature. Comparison of ethanol FE for different CO_2 to ethanol electrolyzers. The comparison is limited to electrolyzers working at total current densities ≥ 100 mA/cm^2 and EtOH FE > 50%

System	Electrolyte	Catalyst	Ethanol FE (%)	Carbon efficiency (%)	E (FC / vs RHE)	J (mA cm^{-2})	Ref.
Flow cell	3M KOH	Im-Mo_3P	77	... < 5*	-0.5 vs RHE	90-120	47
	1M KOH	V-doped Cu_2Se	68	... < 5	-0.9 vs RHE	300	48
	1M KOH	Cu-SC_{12}	57	... < 5	-1.2 vs RHE	~200	49
	1M KOH	Ag@$Cu_{0.05}$	53	... < 5	-1.0 vs RHE	320	50
	1M KOH	K–F–Cu	53	... < 5	-0.7 vs RHE	432	51
MEA	1M KOH	Ag^+-Doped InSe	69	... < 5	3	271	40
	0.02 M H_2SO_4	**Cu-Ag**	**45**	**63**	**3.5**	**200**	**This work**

* 5% is the maximum carbon efficiency for the alkaline electrolyzer.

Supplementary Table 3. Diffusion-coefficients in $m^2 s^{-1}$.[2]

Constant	Value
D_{H^+}	9.311×10^{-9}
D_{Na^+}	1.334×10^{-9}
D_{K^+}	1.957×10^{-9}
D_{Cs^+}	2.06×10^{-9}
D_{OH^-}	5.273×10^{-9}
D_{CO_2}	1.91×10^{-9}
$D_{HCO_3^-}$	1.185×10^{-9}
$D_{CO_3^{2-}}$	0.923×10^{-9}

Supplementary Table 4. Solvation sizes in $m.^2$

Constant	Value
a_{H^+}	0.56×10^{-9}
a_{Na^+}	0.716×10^{-9}
a_{K^+}	0.662×10^{-9}
a_{Cs^+}	0.658×10^{-9}
a_{OH^-}	0.6×10^{-9}
a_{CO_2}	0.23×10^{-9}
$a_{HCO_3^-}$	0.8×10^{-9}
$a_{CO_3^{2-}}$	0.788×10^{-9}

Supplementary Table 5. **Rate constants used for modelling**. [52]

Constant	Value	Units
k_{1f}	3.71×10^{-2}	s^{-1}
k_{1r}	1.99×10^{4}	s^{-1}
k_{2f}	59.99	s^{-1}
k_{2r}	2.28×10^{7}	$mol^{-1}m^{3}s^{-1}$
k_{3f}	2.23	$mol^{-1}m^{3}s^{-1}$
k_{3r}	5.23×10^{5}	s^{-1}
k_{4f}	6.0×10^{6}	$mol^{-1}m^{3}s^{-1}$
k_{4r}	1.07×10^{6}	s^{-1}
k_{wf}	2.4×10^{-2}	$mol\ m^{3}s^{-1}$
k_{wr}	2.4×10^{6}	$mol^{-1}m^{3}s^{-1}$

Supplementary Table 6. **h parameters used to estimate Sechenov's constant** in $m^3 kmol^{-1}$.[53]

Constant	Value
h_{Na^+}	0.1143
h_{K^+}	0.0922
h_{Cs^+}	0.0759
h_{OH^-}	0.0839
$h_{HCO_3^-}$	0.0967
$h_{CO_3^{2-}}$	0.1423
$h_{SO_4^{2-}}$	0.1117
$h^0_{CO_2}$	-0.0172
$h^T_{CO_2}$	-0.00038

Supplementary Table 7. **Contributions to Gibbs free energies of gas species and adsorbates**, including zero-point energy (ZPE), enthalpic temperature correction ($\int C_P \mathrm{d}T$), entropy (S). H_2 refers to the energy used in the computational hydrogen electrode model

Species	$\boldsymbol{ZPE}$ $(\boldsymbol{eV})$	$\int \boldsymbol{C_P}\mathbf{d}\boldsymbol{T}$ $(\boldsymbol{eV})$	$-\boldsymbol{TS}$ (**eV**)	$\boldsymbol{G}-\boldsymbol{E}$ $(\boldsymbol{eV})$
CO_2(g)	0.31	0.10	-0.66	-0.25
H_2O(l)	0.57	0.10	-0.67	0
H_2(g)	0.27	0.08	-0.40	-0.05
$HCCO^*$	0.58	0.11	-0.21	0.48
$HCCOH^*$	0.91	0.09	-0.15	0.85
HCCHO*	0.85	0.11	-0.25	0.71
H_2CCO*	0.88	0.11	-0.20	0.79
H^*	0.17	0.01	-0.01	0.17

Supplementary Table 8. Full cell voltage and RI for MEA electrolyzer at different current densities. Nafion 117 was used as the CEM with 0.01 M H_2SO_4 as anolyte

Current density (mA/cm^2)	50	100	200
Full cell voltage (v)	3.5	3.8	4.6
RI (v)	0.26	0.52	0.93

References:

1. Stern, O. Zur Theorie der Elektrolytischen Doppelschicht. *Zeitschrift fur Elektrochemie* **30**, 508–516 (1924).
2. Bohra, D., Chaudhry, J. H., Burdyny, T., Pidko, E. A. & Smith, W. A. Modeling the Electrical Double Layer to Understand the Reaction Environment in a CO_2 Electrocatalytic System. *Energy Environ. Sci* **12**, 3380 (2019).
3. Johnson, E. F., Boutin, E. & Haussener, S. Surface Charge Boundary Condition Often Misused in CO_2 Reduction Models. *Journal of Physical Chemistry C* **127**, 18784–18790 (2023).
4. Butt, E. N., Padding, J. T. & Hartkamp, R. Size-modified Poisson-Nernst-Planck approach for modeling a local electrode environment in CO_2 electrolysis. *Sustain Energy Fuels* **7**, 144–154 (2022).
5. Paneth-Berlin, H. 5 08 Zeitschrift for Elektrochemie. doi:10.1002/bbpc.192400182.
6. Wang, H. & Pilon, L. Accurate Simulations of Electric Double Layer Capacitance of Ultramicroelectrodes. *Journal of Physical Chemistry C* **115**, 16711–16719 (2011).
7. Booth, F. The Dielectric Constant of Water and the Saturation Effect. *J Chem Phys* **19**, 391–394 (1951).
8. Booth, F. Dielectric Constant of Polar Liquids at High Field Strengths. *J Chem Phys* **23**, 453–457 (1955).
9. Paunov, V. N., Dimova, R. I., Kralchevsky, P. A., Broze, G. & Mehreteab, A. The Hydration Repulsion between Charged Surfaces as an Interplay of Volume Exclusion and Dielectric Saturation Effects. *J Colloid Interface Sci* **182**, 239–248 (1996).

10. Weng, L. C., Bell, A. T. & Weber, A. Z. Modeling Gas-Diffusion Electrodes for CO_2 Reduction. *Physical Chemistry Chemical Physics* **20**, 16973–16984 (2018).

11. Kresse, G. & Hafner, J. Ab Initio Molecular Dynamics for Liquid Metals. *Phys Rev B* **47**, 558–561 (1993).

12. Kresse, G. & Furthmüller, J. Efficiency of Ab-Initio Total Energy Calculations for Metals and Semiconductors Using a Plane-Wave Basis Set. *Comput Mater Sci* **6**, 15–50 (1996).

13. Kresse, G. & Furthmüller, J. Efficient Iterative Schemes for Ab Initio Total-Energy Calculations Using a Plane-Wave Basis Set. *Phys Rev B* **54**, 11169–11186 (1996).

14. Payne, M. C., Teter, M. P., Allan, D. C., Arias, T. A. & Joannopoulos, J. D. Iterative Minimization Techniques for Ab Initio Total-Energy Calculations: Molecular Dynamics and Conjugate Gradients. *Rev Mod Phys* **64**, 1045–1097 (1992).

15. Blöchl, P. E. Projector Augmented-Wave Method. *Phys Rev B* **50**, 17953–17979 (1994).

16. Joubert, D. From Ultrasoft Pseudopotentials to the Projector Augmented-Wave Method. *Phys Rev B Condens Matter Mater Phys* **59**, 1758–1775 (1999).

17. Monkhorst, H. J. & Pack, J. D. Special Points for Brillouin-Zone Integrations. *Phys Rev B* **13**, 5188–5192 (1976).

18. Grimme, S., Antony, J., Ehrlich, S. & Krieg, H. A Consistent and Accurate Ab Initio Parametrization of Density Functional Dispersion Correction (DFT-D) for the 94 Elements H-Pu. *J Chem Phys* **132**, 154104 (2010).

19. Nørskov, J. K. *et al.* Origin of the Overpotential for Oxygen Reduction at a Fuel-Cell Cathode. *J Phys Chem B* **108**, 17886–17892 (2004).

20. Bader, R. F. W. A Quantum Theory of Molecular Structure and its Applications. *Chem Rev* **91**, 893–928 (1991).

21. Tang, W., Sanville, E. & Henkelman, G. A grid-based Bader Analysis Algorithm without Lattice Bias. *J. Phys. Condens. Matter* **21**, 084204 (2009).

22. Montoya, J. H., Shi, C., Chan, K. & Nørskov, J. K. Theoretical Insights into a CO Dimerization Mechanism in CO_2 Electroreduction. *J Phys Chem Lett* **6**, 2032–2037 (2015).

23. Nosé, S. A Unified Formulation of the Constant Temperature Molecular Dynamics Methods. *J Chem Phys* **81**, 511–519 (1984).

24. Hoover, W. G. Canonical Dynamics: Equilibrium Phase-Space Distributions. *Phys Rev A (Coll Park)* **31**, 1695–1697 (1985).

25. Nosé, S. Constant Temperature Molecular Dynamics Methods. *Progress of Theoretical Physics Supplement* **103**, 1–46 (1991).

26. Xie, Y. *et al.* High Carbon Utilization in CO_2 Reduction to Multi-Carbon Products in Acidic Media. *Nat Catal* **5**, 564–570 (2022).

27. O'Brien, C. P. *et al.* Single Pass CO_2 Conversion Exceeding 85% in the Electrosynthesis of Multicarbon Products via Local CO_2 Regeneration. *ACS Energy Lett* **6**, 2952–2959 (2021).

28. Xu, Y. *et al.* A Microchanneled Solid Electrolyte for Carbon-Efficient CO_2 Electrolysis. *Joule* **6**, 1333–1343 (2022).

29. Yang, K. *et al.* Cation-Driven Increases of CO_2 Utilization in a Bipolar Membrane Electrode Assembly for CO_2 Electrolysis. *ACS Energy Lett* **6**, 4291–4298 (2021).

30. Huang, J. E. *et al.* CO_2 Electrolysis to Multicarbon Products in Strong Acid. *Science (1979)* **372**, 1074–1078 (2021).

31. Zhao, Y. *et al.* Conversion of CO_2 to Multicarbon Products in Strong Acid by Controlling the Catalyst Microenvironment. *Nat Synth* **2**, 403–412 (2023).

32. Spurgeon, J. M. & Kumar, B. A Comparative Technoeconomic Analysis of Pathways for Commercial Electrochemical CO_2 Reduction to Liquid Products. *Energy Environ Sci* **11**, 1536–1551 (2018).

33. Masel, R. I. *et al.* An Industrial Perspective on Catalysts for Low-Temperature CO_2 Electrolysis. *Nat Nanotechnol* **16**, 118–128 (2021).

34. Zhao, Y. *et al.* Conversion of CO_2 to Multicarbon Products in Strong Acid by Controlling The Catalyst Microenvironment. *Nat Synth 2023* 1–10 (2023).

35. Huang, J. E. *et al.* CO_2 Electrolysis to Multicarbon Products in Strong Acid. *Science (1979)* **372**, 1074–1078 (2021).

36. Hansen, K. U., Cherniack, L. H. & Jiao, F. Voltage Loss Diagnosis in CO_2 Electrolyzers Using Five-Electrode Technique. *ACS Energy Lett* **7**, 4504–4511 (2022).

37. Montoya, J. H., Shi, C., Chan, K. & Nørskov, J. K. Theoretical Insights into a CO Dimerization Mechanism in CO_2 Electroreduction. *Journal of Physical Chemistry Letters* **6**, 2032–2037 (2015).

38. Adnan, M. A. *et al.* Directly-Deposited Ultrathin Solid Polymer Electrolyte for Enhanced CO_2 Electrolysis. *Adv Energy Mater* 13, 2203158 (2023).

39. Shang, L. *et al.* Efficient CO_2 Electroreduction to Ethanol by Cu_3Sn Catalyst. *Small Methods* **6**, 2101334 (2022).

40. Wang, X. *et al.* Ag^+-Doped InSe Nanosheets for Membrane Electrode Assembly Electrolyzer toward Large-Current Electroreduction of CO_2 to Ethanol. *Angew Chem Int Ed* **62**, e202313646 (2023).

41. Gabardo, C. M. *et al.* Continuous Carbon Dioxide Electroreduction to Concentrated Multi-carbon Products Using a Membrane Electrode Assembly. *Joule* **3**, 2777–2791 (2019).

42. Li, F. *et al.* Cooperative CO_2-to-Ethanol Conversion via Enriched Intermediates at Molecule–Metal Catalyst Interfaces. *Nat Catal 2019 3:1* **3**, 75–82 (2019).

43. Gu, Z. *et al.* Efficient Electrocatalytic CO_2 Reduction to C_{2+} Alcohols at Defect-Site-Rich Cu Surface. *Joule* **5**, 429–440 (2021).

44. Wang, X. *et al.* Efficient electrically powered CO_2-to-Ethanol via Suppression of Deoxygenation. *Nat Energy* **5**, 478–486 (2020).

45. Miao, R. K. *et al.* Electroosmotic Flow Steers Neutral Products and Enables Concentrated Ethanol Electroproduction from CO_2. *Joule* **5**, 2742–2753 (2021).

46. O'Brien, C. P. *et al.* Single Pass CO_2 Conversion Exceeding 85% in the Electrosynthesis of Multicarbon Products via Local CO_2 Regeneration. *ACS Energy Lett* **6**, 2952–2959 (2021).

47. Esmaeilirad, M. *et al.* Efficient Electrocatalytic Conversion of CO_2 to Ethanol Enabled by Imidazolium-Functionalized Ionomer Confined Molybdenum Phosphide. *Appl Catal B* **317**, 121681 (2022).

48. Sun, W. *et al.* V-Doped Cu_2Se Hierarchical Nanotubes Enabling Flow-Cell CO_2 Electroreduction to Ethanol with High Efficiency and Selectivity. *Adv Mater* **34**, 2207691 (2022).

49. Lin, Y. *et al.* Tunable CO_2 Electroreduction to Ethanol and Ethylene with Controllable Interfacial Wettability. *Nat Commun 2023 14:1* **14**, 1–12 (2023).

50. Cai, Z. *et al.* Hierarchical Ag-Cu Interfaces Promote C-C Coupling in Tandem CO_2 Electroreduction. *Appl Catal B* **325**, 122310 (2023).

51. Peng, C. *et al.* Surface Co-Modification of Halide Anions and Potassium Cations Promotes High-Rate CO_2-to-Ethanol Electrosynthesis. *Adv Mater.* **34**, 2204476 (2022).

52. Bohra, D., Chaudhry, J. H., Burdyny, T., Pidko, E. A. & Smith, W. A. Modeling the Electrical Double Layer to Understand the Reaction Environment in a CO_2 Electrocatalytic System. *Energy Environ Sci* **12**, 3380–3389 (2019).

53. Chung, D. W., Ebner, M., Ely, D. R., Wood, V. & Edwin García, R. Validity of the Bruggeman Relation for Porous Electrodes. *Model Simul Mat Sci Eng* **21**, (2013).